# CHAPTER-3 FUNCTIONS

## 3.1. Review of Relations and Functions

## Cartesian Product

Given sets $A=\{3,4\}$ and $B=\{2,4,5\}$. Then, the set $\{(3,2),(3,4),(3,5),(4,2),(4,4),(4,5)\}$ is the Cartesian product of $A$ and $B$, and it is denoted by $A\times B$.

**Definition 3.1:** Suppose $A$ and $B$ are sets. The Cartesian product of $A$ and $B$, denoted by $A\times B$, is the set which contains every ordered pair whose first coordinate is an element of $A$ and second coordinate is an element of $B$, i.e.

$$A\times B=\{(a,b):a\in A \text{ and } b\in B\}.$$

**Example3.1** For $A=\{2,4\}$ and $B=\{-1,3\}$, we have

a) $AXB=\{(2,-1),(2,3),(4,-1),(4,3)\}$, and

b) $BXA=\{(-1,2),(-1,4),(3,2),(3,4)\}$.

From this example, we can see that $A \times B$ and $B \times A$ are not equal. Recall that two sets are equal if one is a subset of the other and vice versa. To check equality of Cartesian products we need to define equality of ordered pairs.

**Definition 3.2: (Equality of ordered Pairs)**

Two ordered pairs $(a, b)$ and $(c, d)$ are equal if and only if $a = c$ and $b = d$.

**Example 3.2**: Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$. Then,

$$A \times B = \{(1,a),(1,b),(1,c),(2,a),(2,b),(2,c),(3,a),(3,b),(3,c)\}.$$

**Definition 3.3: (Relation)**

If $A$ and $B$ are sets, any subset of $A \times B$ is called a relation from $A$ into $B$.

Suppose $R$ is a relation from a set $A$ to a set $B$. Then, $R \subseteq A \times B$ and hence for each $(a,b) \in A \times B$, we have either $(a,b) \in R$ or $(a,b) \notin R$. If $(a,b) \in R$, we say "$a$ is $R$-related (or simply related) to $b$", and write $aRb$. If $(a,b) \notin R$, we say that "$a$ is not related to $b$". In particular if $R$ is a relation from a set $A$ to itself, then we say that $R$ is a relation on $A$.

**Example 3.3**:

1. Let $A = \{1,3,5,7\}$ and $B = \{6,8\}$. Let $R$ be the relation "less than" from $A$ to $B$. Then,
   $R = \{(1,6),(1,8),((3,6),(3,8),(5,6),(5,8),(7,8)\}$.
2. Let $A = \{1,2,3,4,5\}$ and $B = \{a,b,c\}$.

a) The following are relations from $A$ into $B$;

i) $R_1 = \{(1,a)\}$

ii) $R_2 = \{(2,b),(3,b),(4,c),(5,a)\}$

iii) $R_3 = \{(1,a),((2,b),(3,c)\}$

b) The following are relations from $B$ to $A$;

i) $R_4 = \{(a,3),(b,1)\}$

ii) $R_5 = \{(b,2),(c,4),(a,2),(b,3)\}$

iii) $R_6 = \{(b,5)\}$

**Definition 3.4:** Let R be a relation from A into B. Then,

a) The domain of R, denoted by Dom(R) , is the set of first coordinates of the elements of R , i.e

$$Dom(R) = \{a \in A : (a,b) \in R\}$$

b) The range of R , denoted by Range(R) , is the set of second coordinates of elements of R , i.e

$$Range(R) = \{b \in B : (a,b) \in R\}$$

**Remark**: If R is a relation form the set A to the set B , then the set B is called the **codomain of the relation**. The range of relation is always a subset of the codomain.

**Example 3.4**: 1) $R = \{(4,7), (5,8), (6,10)\}$ is a relation from $A = \{1, 2, 3, 4, 5, 6\}$ to $B = \{6, 7, 8, 9, 10\}$. The Dom(R) $= \{4, 5, 6\}$, Range(R)$= \{7, 8, 10\}$ and

Codom(R)$= \mathrm{B} = \{6, 7, 8, 9, 10\}$.

2) $R = \{(8,2), (6, -3), (5, 7), (5, -3)\}$ is a relation from $\{5, 6, 8\}$ and $\{2, -3, 7\}$, where $\{5, 6, 8\}$ is the domain and $\{2, -3, 7\}$ is the range.

**Remark:**

- If $(a, b) \in R$ for a relation $R$ , we say $a$ is related to (or paired with) b. Note that $a$ may also be paired with an element different from b. In any case, b is called the **image** of $a$ while $a$ is called the **pre-image** of b.
- If the domain and/or range of a relation is infinite, we cannot list each element assignment, so instead we use set builder notation to describe the relation.
- The situation we will encounter most frequently is that of a relation defined by an equation or formula. For example, $\mathrm{R} = \{(\mathrm{x}, \mathrm{y}): \mathrm{y} = 2\mathrm{x} - 3,\ \mathrm{x},\ \mathrm{y} \in \mathbb{R}\}$ is a relation for which the range value is 3 less than twice the domain value. Hence, $(0, -3), (0.5, -2)\ and\ (-2, -7)$ are examples of ordered pairs that are of the assignment.

**Example 3.5**:

1. Let $A = \{1,2,3,4,6\}$

   Let $R$ be the relation on $A$ defined by $R = \{(a,b) : a,b \in A, a \text{ is a factor of } b\}$. Find the domain and range of $R$.

2. Let $A = \{1,2,3,4,5\}$ and $B = \{1,2,3,\cdots,67\}$.

   Let $R = \{(x,y) \in A \times B : x \text{ is cube root of } y\}$. Find a) $R$ b) $Dom(R)$ c) $Range(R)$

**Solution**: We have

$$R = \{(1,1),(1,2),(1,3),(1,4),(1,6),(2,2),(2,4),(2,6),(3,3),(3,6),(4,4),(6,6)\}.$$

Then, $Dom(R) = \{1,2,3,4,6\}$ and $Range(R) = \{1,2,3,4,6\}$.

**Solution**: We have $1 = \sqrt[3]{1}, 2 = \sqrt[3]{8}, 3 = \sqrt[3]{27}, 4 = \sqrt[3]{64}, 5 = \sqrt[3]{125}$ and 1,8,27 and 64 are in $B$ whereas 125 is not in $B$. Thus $R = \{(1, 1), (2,8), (3, 27), (4, 64)\}$, $Dom(R) = \{1, 2, 3, 4\}$ and $Range(R) = \{1, 8, 27, 64\}$

**Remark**:

1. A relation $R$ on a set $A$ is called

   i) a universal relation if $R = A \times A$

   ii) identity relation if $R = \{(a,a) : a \in A\}$

   iii) void or empty relation if $R = \phi$

2. If $R$ is a relation from $A$ to $B$, then the inverse relation of $R$, denoted by $R^{-1}$, is a relation from $B$ to $A$ and is defined as:

$R^{-1} = \{(y,x) : (x,y) \in R\}$.

Observe that $Dom(R) = Range(R^{-1})$ and $Range(R) = Dom(R^{-1})$. For instance, if $R = \{(1,4),(9,15),(10,2)\}$ is a relation on a set $A = \{1,2,3,\cdots,20\}$, then $R^{-1} = \{(4,1),(15,9),(2,10)\}$

**Example 3.6**: Let $R$ be a relation defined on $IN$ by $R = \{(a,b) : a,b \in IN,\ a+2b = 11\}$.

Find a) $R$ b) $Dom(R)$ c) $Range(R)$ d) $R^{-1}$

**Solution**: The smallest natural number is 1.

$b = 1 \Rightarrow a + 2(1) = 11 \Rightarrow a = 9$

$b = 2 \Rightarrow a + 2(2) = 11 \Rightarrow a = 7$

$b = 3 \Rightarrow a + 2(3) = 11 \Rightarrow a = 5$

$b = 4 \Rightarrow a + 2(4) = 11 \Rightarrow a = 3$

$b = 5 \Rightarrow a + 2(5) = 11 \Rightarrow a = 1$

$b = 6 \Rightarrow a + 2(6) = 11 \Rightarrow a = -1 \notin IN$

Therefore, $R = \{(9,1),(7,2),(5,3),(3,4),(1,5)\}$, $Dom(R) = \{1,3,5,7,9\}$, $Range(R) = \{1,2,3,4,5\}$ and $R^{-1} = \{(1,9),(2,7),(3,5),(4,3),(5,1)\}$.

# Functions

Mathematically, it is important for us to distinguish among the relations that assign a unique range element to each domain element and those that do not.

**Definition 3.5**: (**Function**) A function is a relation in which each element of the domain corresponds to exactly one element of the range.

**Example 3.7**: Determine whether the following relations are functions.

a) $\{(5,-2),(3,5),(3,7)\}$ b) $\{(2,4),(3,4),(6,-4)\}$

**Solution**:

a) Since the domain element 3 is assigned to two different values in the range, 5 and 7, it is not a function.

b) Each element in the domain, {2, 3, 6}, is assigned no more than one value in the range, 2 is assigned only 4, 3 is assigned only 4, and 6 is assigned only – 4. Therefore, it is a function.

**Remark**: Map or mapping, transformation and correspondence are synonyms for the word function. If $f$ is a function and $(x, y) \in f$, we say *x* is mapped to *y*.

**Definition 3.6**: A relation $f$ from $A$ into $B$ is called a function from $A$ into $B$, denoted by

$$f : A \to B \quad \text{or} \quad A \xrightarrow{f} B$$

if and only if

(i) $Dom(f) = A$

(ii) No element of $A$ is mapped by $f$ to more than one element in $B$, i.e. if $(x, y) \in f$ and $(x, z) \in f$, then $y = z$.

Remark: 1. If to the element $x$ of A corresponds $y(\in B)$ under the function $f$, then we write $f(x) = y$ and $y$ is called the image of $x$ under $f$ and x is called a pre-image of $y$ under $f$.

2. The symbol $f(x)$ read as "$f$ of $x$" but not "$f$ times $x$".

3. In order to show that a relation $f$ from A into B is a function, we first show that the domain of $f$ is A and next we show $f$ is well defined or single-valued, i.e. if $x =$y in A, then $f(x) = f(y)$ in B for all $x, y \in A$.

**Example 3.8:**

1. Let $A=\{1,2,3,4\}$ and $B=\{1,6,8,11,15\}$. Which of the following are functions from $A$ to $B$.

   a) $f$ defined by $f(1)=1, f(2)=6, f(3)=8, f(4)=8$

   b) $f$ defined by $f(1)=1, f(2)=6, f(3)=15$

   c) $f$ defined by $f(1)=6, f(2)=6, f(3)=6, f(4)=6$

   d) $f$ defined by $f(1)=1, f(2)=6, f(2)=8, f(3)=8, f(4)=11$

   e) $f$ defined by $f(1)=1, f(2)=8, f(3)=11, f(4)=15$

**Solution**:

a) $f$ is a function because to each element of $A$ there corresponds exactly one element of $B$.

b) $f$ is a function each element of $A$ there corresponds exactly one element of $B$.

c) $f$ is a function because to each element of $A$ there corresponds exactly one element of $B$. In the given function, the images of all element of $A$ are the same.

d) $f$ is not a function because there are two elements of $B$ which are corresponding to 2. In other words, the image of 2 is not unique.

e) $f$ is a function because to each element of $A$ there corresponds exactly one element of $B$.

As with relations, we can describe a function with an equation. For example, $y = 2x + 1$ is a function, since each $x$ will produce only one.

2) Let $f = \{(x, y): y = x^2\}$. Then $f$ maps:

| | |
|---|---|
| 1 to 1 | -1 to 1 |
| 2 to 4 | -2 to 4 |
| 3 to 9 | -3 to 9 |

More generally any real number $x$ is mapped to its square. As the square of a number is unique, $f$ maps every real number to a unique number. Thus $f$ is a function from $\mathbb{R}$ into $\mathbb{R}$.

**Note that:** The independent variable refers to the variable representing possible values in the domain, and the dependent variable refers to the variable representing possible values in the range. Thus, in our usual ordered pair notation $(x, y)$, $x$ is the independent variable and y is the dependent variable.

# Domain, Codomain and range of a function

For the function $f : A \to B$

(i) The set $A$ is called the domain of $f$

(ii) The set $B$ is called the codomain of $f$

(iii) The set $\{f(x) : x \in A\}$ of all image of elements of $A$ is called the range of $f$

**Example 3.9**:

1. Let $A = \{1,2,3\}$ and $B = \{1,2,3,\cdots,10\}$. Let $f : A \to B$ be the correspondence which assigns to each element in $A$, its square. Thus, we have $f(1) = 1,\ f(2) = 4,\ f(3) = 9$. Therefore, $f$ is a function and $Dom(f) = \{1,2,3\}$, $Range(f) = \{1,4,9\}$ and codomain of $f$ is $\{1,2,3,\cdots,10\}$.

2. Let $A = \{2,4,6,7,9\}, B = IN$. Let $x$ and $y$ represent the elements in the sets $A$ and $B$, respectively. Let $f : A \to B$ be a function defined by $f(x) = 15x + 17, x \in A$.

The variable $x$ can take values 2, 4, 6, 7, 9. Thus, we have

$$f(2) = 15(2) + 17 = 47,\ f(4) = 77,\ f(6) = 107,\ f(7) = 122,\ f(9) = 152.$$

This implies that $Dom(f) = \{2,4,6,7,9\}$, $Range(f) = \{47,77,107,122,152\}$ and codomain of $f$ is $IN$.

3. Let $f$ be the subset of $\mathbb{Q} \times \mathbb{Z}$ defined by $f = \left\{\left(\frac{p}{q},\ p\right) : p, q \in \mathbb{Z}, q \neq 0\right\}$. Is $f$ a function?

**Solution**: First we note that $\mathrm{Dom}(f) = \mathbb{Q}$ . Then $f$ satisfies condition (i) in the definition of a function. Now, $\left(\frac{2}{3},\ 2\right), \left(\frac{4}{6},\ 4\right) \in f$ but, $f\left(\frac{2}{3}\right) = 2 \neq 4 = f\left(\frac{4}{6}\right)$. Thus $f$ is not well defined. Hence, $f$ is not a function from $\mathbb{Q}$ to $\mathbb{Z}$ .

4. Let $f$ be the subset of $\mathbb{Z} \times \mathbb{Z}$ defined by $f = \{(mn, m+n) : m, n \in \mathbb{Z}\}$ . Is $f$ a function?

**Solution**: First we show that $f$ satisfies condition (i) in the definition. Let $x$ be any element of $\mathbb{Z}$. Then $x = x.1$. Hence $(x,\ x+1) = (x*1,\ x+1)$. This implies that $x \in Dom(f)$. Thus, $\mathbb{Z} \subseteq Dom(f)$. However, $Dom(f) \subseteq \mathbb{Z}$ and so, $Dom(f) = \mathbb{Z}$. Now, $4 \in \mathbb{Z}$ and $4 = 4*1 = 2*2$. Thus $(4*1, 4+1)$ and $(2*2, 2+2)$ are in $f$. Hence we find that $4*1 = 2*2$ and $f(4+1) = 5 \neq 4 = f(2*2)$. This implies that $f$ is not well defined, i.e. $f$ does not satisfy condition (ii). Hence, $f$ is not a function from $\mathbb{Z}$ to $\mathbb{Z}$.

# Exercise

5. Determine whether the following equations determine $y$ as a function of $x$, if so, find the domain.

$a)\ y = -3x + 5$ b) $y = \frac{2x}{3x-5}$ c) $y^2 = x$

$$d) y = \sqrt{3x - x^2}.$$

**Solution:** a) To determine whether $y = -3x + 5$ gives $y$ as a function of $x$ , we need to know whether each x-value uniquely determines a y-value. Looking at the equation $y = -3x + 5$ , we can see that once is chosen we multiply it by – 3 and then add 5. Thus, for each x there is a unique $y$ . Therefore $y = -3x + 5$ is a function.

b) Looking at the equation $y = \frac{2x}{3x-5}$ carefully, we can see that each x-value uniquely determines a y-value (one x-value can not produce two different y-values). Therefore $y = \frac{2x}{3x-5}$ is a function.

As for its domain, we ask ourselves. Are there any values of that must be excluded? Since $y = \frac{2x}{3x-5}$ is a fractional expression, we must exclude any value of that makes the denominator equal to zero. We must have $3x - 5 \neq 0 \Leftrightarrow x \neq \frac{5}{3}$.

Therefore the domain consists of all real numbers except for $\frac{5}{3}$. Thus $Dom(f) = \{x: x \neq \frac{5}{3}\}$.

c) For the equation $y^2 = x$, if we choose $x = 9$ we get , $y^2 = 9$ which gives $y = \pm 3$. In other words, there are two $y$-values associated with $x = 9$. Therefore $y^2 = x$ is not a function.

6) Find the domain of the function

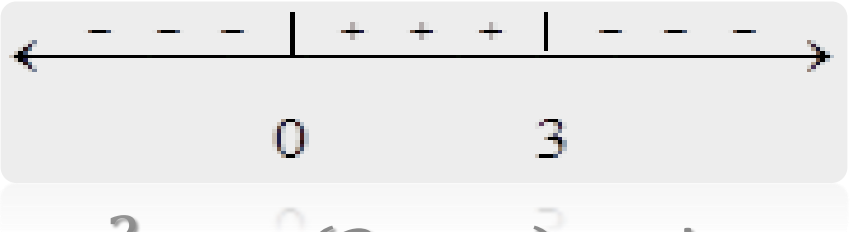
$y = \sqrt{3x - x^2}$.

**Solution**: Since $y$ is defined and real when the expression under the radical is non-negative, we need to satisfy the inequality

$$3x - x^2 \geq 0 \Leftrightarrow x(3 - x) \geq 0$$

This is a quadratic inequality, which can be solved by analyzing signs:

Sign of $3x - x^2$




Since we want $3x - x^2 = x(3 - x)$ to be non-negative, the sign analysis shows us that the domain is $\{x: 0 \leq x \leq 3\}$or [0, 3].

# 3.2 Real Valued functions and their properties

Let $f$ be a function from set $A$ to set $B$. If $B$ is a subset of real number system $\Re$, then $f$ is called a real valued function, and in particular if $A$ is also a subset of $\Re$, then $f : A \to B$ is called a real function.

**Example 3.10**: 1. The function $f : \Re \to \Re$ defined by $f(x) = x^2 + 3x + 7$, $x \in \Re$ is a real function.

2. The function $f : \Re \to \Re$ defined as $f(x) = |x|$ is also a real valued function.

## Operations on functions

*Functions are not numbers. But just as two numbers $a$ and $b$ can be added to produce a new number $a + b$, so two functions $f$ and $g$ can be added to produce a new function $f + g$. This is just one of the several operations on functions that we will describe in this section.*

Consider functions $f$ and $g$ with formulas $f(x)=\frac{x-3}{2}$, $g(x)=\sqrt{x}$. We can make a new function $f+g$ by having it assign to $x$ the value $\frac{x-3}{2}+\sqrt{x}$, that is,

$$(f+g)(x)=f(x)+g(x)=\frac{x-3}{2}+\sqrt{x}\ .$$

**Definition 3.7: Sum, Difference, Product and Quotient of two functions**

Let $f(x)$ and $g(x)$ be two functions. We define the following four functions:

1. $(f+g)(x)=f(x)+g(x)$ — The sum of the two functions
2. $(f-g)(x)=f(x)-g(x)$ — The difference of the two functions
3. $(f\cdot g)(x)=f(x)g(x)$ — The product of the two functions
4. $\left(\frac{f}{g}\right)(x)=\frac{f(x)}{g(x)}$ — The quotient of the two functions (provided $g(x)\neq 0$)

Since an $x-$value must be an input into both $f$ and $g$, the domain of $(f+g)(x)$ is the set of all $x$ common to the domain of $f$ and $g$. This is usually written as $Dom(f+g)=Dom(f)\cap Dom(g)$. Similar statements hold for the domains of the difference and product of two functions. In the case of the quotient, we must impose the additional restriction that all elements in the domain of $g$ for which $g(x)=0$ are excluded.

**Example 3.11**:

1. Let $f(x)=3x^2+2$ and $g(x)=5x-4$. Find each of the following and its domain

a) $(f+g)(x)$ b) $(f-g)(x)$ c) $(f.g)(x)$ d) $\left(\frac{f}{g}\right)(x)$

**Solution**:

a) $(f+g)(x)=f(x)+g(x)=(3x^2+2)+(5x-4)=3x^2+5x-2$

b) $(f-g)(x)=f(x)-g(x)=(3x^2+2)-(5x-4)=3x^2-5x+6$

c) $(f\cdot g)(x)=(3x^2+2)(5x-4)=15x^3-12x^2+10x-8$

d) $\left(\frac{f}{g}\right)(x)=\frac{f(x)}{g(x)}=\frac{3x^2+2}{5x-4}$

We have

$$Dom(f+g)=Dom(f-g)=Dom(fg)=Dom(f)\cap Dom(g)=\Re\cap\Re=\Re$$

$$Dom\left(\frac{f}{g}\right)=\left[Dom(f)\cap Dom(g)\right]\setminus\{x:g(x)=0\}=\Re\setminus\left\{\frac{5}{4}\right\}$$

2. Let $f(x)=\sqrt[4]{x+1}$ and $g(x)=\sqrt{9-x^2}$, with respective domains $[-1,\infty)$ and $[-3,3]$. Find formulas for $f+g,\ f-g,\ f\cdot g,\frac{f}{g}$ and $f^3$ and give their domains.

**Solution**:

| Formula | Domain |
|---|---|
| $(f+g)(x) = f(x)+g(x) = \sqrt[4]{x+1}+\sqrt{9-x^2}$ | $[-1,3]$ |
| $(f-g)(x) = f(x)-g(x) = \sqrt[4]{x+1}-\sqrt{9-x^2}$ | $[-1,3]$ |
| $(f\cdot g)(x) = f(x)\cdot g(x) = \sqrt[4]{x+1}\cdot\sqrt{9-x^2}$ | $[-1,3]$ |
| $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{\sqrt[4]{x+1}}{\sqrt{9-x^2}}$ | $[-1,3)$ |
| $f^3(x) = \left(f(x)\right)^3 = \left(\sqrt[4]{x+1}\right)^3 = (x+1)^{\frac{3}{4}}$ | $[-1,\infty)$ |

There is yet another way of producing a new function from two given functions.

**Definition 3.8: (Composition of functions)**

Given two functions $f(x)$ and $g(x)$, the composition of the two functions is denoted by $f \circ g$ and is defined by:

$$(f \circ g)(x) = f[g(x)].$$

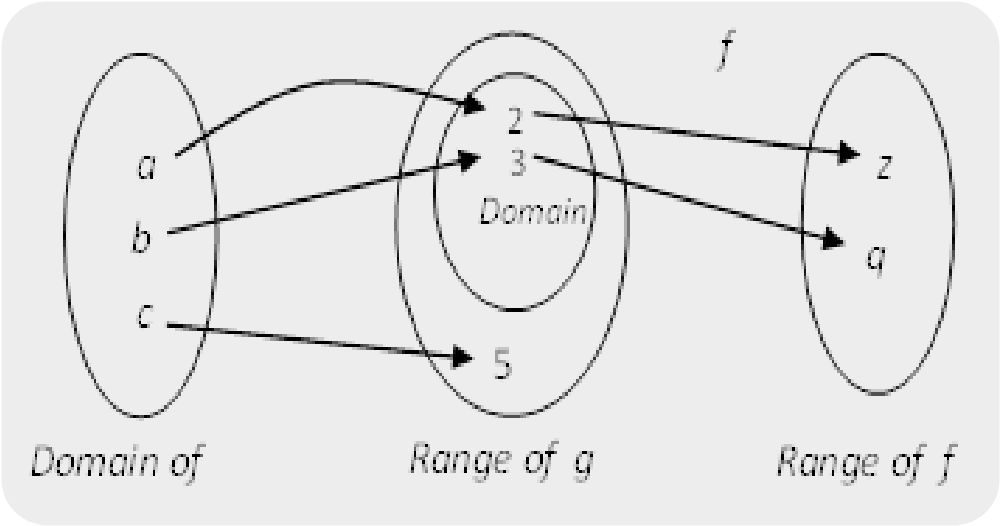
$(f \circ g)(x)$ is read as "$f$ composed with $g$ of $x$". The domain of $f \circ g$ consists of those $x's$ in the domain of $g$ whose range values are in the domain of $f$, i.e. those $x's$ for which $g(x)$ is in the domain of $f$.

**Example 3.12**:

1. Suppose $f = \{(2,z),(3,q)\}$ and $g = \{(a,2),(b,3),(c,5)\}$. The function $(f \circ g)(x) = f(g(x))$ is found by taking elements in the domain of $g$ and evaluating as follows:

   $(f \circ g)(a) = f(g(a)) = f(2) = z, \quad (f \circ g)(b) = f(g(b)) = f(3) = q$

If we attempt to find $f(g(c))$ we get $f(5)$, but 5 is not in the domain of $f(x)$ and so we cannot find $(f \circ g)(c)$. Hence, $f \circ g = \{(a,z),(b,q)\}$. The figure below illustrates this situation.




2. Given $f(x) = 5x^2 - 3x + 2$ and $g(x) = 4x + 3$ , find

*a)* $(f \circ g)(-2)$ b) $(g \circ f)(2)$ c) $(f \circ g)(x)$ d) $(g \circ f)(x)$

**Sol:** a) $(f \circ g)(-2) = f\big(g(-2)\big), but\ \ g(-2) = 4(-2) + 3 = -5$

Thus $(f \circ g)(-2) = f\big(g(-2)\big) = f(-5) = 5(-5)^2 - 3(-5) + 2 = 142.$

b) $(g \circ f)(2) = g\big(f(2)\big)$

$= g\big(5(2)^2 - 3(2) + 2\big) = g(16) = 4(16) + 3 = 67.$

c) $(f \circ g)(x) = f\big(g(x)\big) = f(4x + 3)$

$= 5(4x + 3)^2 - 3(4x + 3) + 2 = 80x^2 + 108x + 38$

d) $(g \circ f)(x) = g\big(f(x)\big) = 4\big(5x^2 - 3x + 2\big) + 3 = 20x^2 - 12x + 11$

Note: The domain of a composite function $f(g(x))$ is the **set of those $x$ in the domain of $g$** for which $g(x)$ is in the domain of $f$.

**Example:** Find the domain of $(fog)(x)$

Where $f(x) = \frac{5}{x-1}$ and $g(x) = \frac{4}{3x-2}$

**Solution:** The domain of $g(x)$ consists of all real numbers except $x = \frac{2}{3}$, since this value would cause us to divide by $0$. Likewise, the domain of $f$ consists of all real numbers except $1$. So we need to exclude from the domain of $g(x)$ that value of $x$ for which $g(x) = 1$.

$$\frac{4}{3x-2} = 1 \qquad \text{Set } g(x) \text{ equal to 1}$$

$$4 = 3x - 2 \qquad \text{Multiply by } 3x - 2$$

$$6 = 3x \qquad \text{Adding 2 to both sides}$$

$$x = 2 \qquad \text{Divide by 3}$$

So the domain of $f \circ g$ is the set of all real numbers except $\frac{2}{3}$ and 2. This means that $x \neq \frac{2}{3}$ or $x \neq 2$. So, $Dom\ fog = \left(-\infty, \frac{2}{3}\right) \cup \left(\frac{2}{3},\ 2\right) \cup (2, \infty)$

We now explore the meaning of equality of two functions. Let $f : A \to B$ and $g : A \to B$ be two functions. Then, $f$ and $g$ are subsets of $A \times B$. Suppose $f = g$. Let $x$ be any element of $A$. Then, $(x, f(x)) \in f = g$ and thus $(x, f(x)) \in g$. Since $g$ is a function and $(x, f(x))$, $(x, g(x)) \in g$, we must have $f(x) = g(x)$. Conversely, assume that $g(x) = f(x)$ for all $x \in A$. Let $(x, y) \in f$. Then, $y = f(x) = g(x)$. Thus, $(x, y) \in g$, which implies that $f \subseteq g$. Similarly, we can show that $g \subseteq f$. It now follows that $f = g$. Thus two functions $f : A \to B$ and $g : A \to B$ are equal if and only if $f(x) = g(x)$ for all $x \in A$. In general we have the following definition.

**Definition 3.9: (Equality of functions)**

Two functions are said to be equal if and only if the following two conditions hold:

i) The functions have the same domain;

ii) Their functional values are equal at each element of the domain.

**Example 3.13**:

1. Let $f : Z \to Z^+ \bigcup \{0\}$ and $g : Z \to Z^+ \bigcup \{0\}$ be defined by $f = \{(n, n^2) : n \in Z\}$ and $g = \{(n, |n|^2) : n \in Z\}$. Now, for all $n \in Z$, $f(n) = n^2 = |n|^2 = g(n)$. Thus, $f = g$.

2. Let $f(x) = \dfrac{x^2 - 25}{x - 5}$, $x \in \Re \setminus \{5\}$, and $g(x) = x + 5$, $x \in \Re$. The function $f$ and $g$ are not equal because $Dom(f) \neq Dom(g)$.

# Types of functions and inverse of a function

## One to One functions

**Definition 3.10:** A function $f: A \longrightarrow B$ is called one to one, often written $1-1$, if and only if for all $x, y \in A$, $f(x) = f(y)$, implies $x = y$. OR, If no two elements of A are mapped to one element of B.

**Example 3.14:**

1. If we consider the sets $A = \{1,2,3,\cdots,6\}$ and $B = \{7,a,b,c,d,8,e\}$ and if $f = \{(1,7), (2,a), (3,b), (4,b), (5,c), (6,8)\}$ and $g = \{(1,7), (2,a), (3,b), (4,c), (5,8), (6,d)\}$, then both $f$ and $g$ are functions from $A$ into $B$. Observe that $f$ is not a $1-1$ function because $f(3) = f(4)$ but $3 \neq 4$. However, $g$ is a $1-1$ function.

2. Let $A = \{1,2,3,4\}$ and $B = \{1,4,7,8\}$. Consider the functions
   - i) $f: A \to B$ defined as $f(1) = 1,\ f(2) = 4,\ f(3) = 4,\ f(4) = 8$
   - ii) $g: A \to B$ defined as $g(1) = 4,\ g(2) = 7,\ g(3) = 1,\ g(4) = 8$

   Then, $f$ is not $1-1$, but $g$ is a $1-1$ function.

# Onto functions

**Definition 3.11:** Let $f$ be a function from a set A into a set B. Then $f$ is called an **onto function(or $f$ maps from A onto B)** if every element of B is image of some element in A, i.e, $Range(f) = B$.

**Example 3.15**: 1) Let $A = \{1, 2, 3\}$ and $B = \{1, 4, 5\}$ . The function $f: A \longrightarrow B$ defined as $f(1) = 1, f(2) = 5$ and $f(3) = 1$ is not onto because there is no element in A, whose image under $f$ is 4. The function g: $A \longrightarrow B$ given by $g = \{(1,4), (2, 5), (3, 1), \}$ is onto because each element of B is the image of one element of A .

Note that if A is a non-empty set, the function $i_A: A \longrightarrow A$ defined by $i_A(x) = x$ for all $x \in A$ is a 1 – 1 function from A onto A . $i_A$ is called the **identity map** on A.

2) Consider the relation $f$ from $\mathbb{Z}$ into $\mathbb{Z}$ defined by $f(n) = n^2$ for all $n \in \mathbb{Z}$. Now, domain of $f$ is $\mathbb{Z}$. Also, if $n = n'$, then $n^2 = (n')^2$, i.e. $f(n) = f(n')$. Hence, $f$ is well defined and a function. However, $f(1) = 1 = f(-1)$ and $1 \neq -1$, which implies that $f$ is not 1 – 1. For all $n \in \mathbb{Z}$, $f(n)$ is a non-negative integer. This shows that a negative integer has no pre-image. Hence $f$ is not onto.

3) Consider the relation $f$ from $\mathbb{Z}$ into $\mathbb{Z}$ defined by $f(n) = 2n$ for all $n \in \mathbb{Z}$. As in the previous example, we can show that $f$ is a function. Let $n, n' \in \mathbb{Z}$ and suppose that $f(n) = f(n')$. Then $2n = 2n'$ and thus $n = n'$. Hence, $f$ is 1 – 1. Since for all $n \in \mathbb{Z}$, $f(n)$ is an even integer; we see that an odd integer has no pre-image. Therefore $f$ is not onto.

# 1 – 1 Correspondence

**Definition 3.12:** A function $f: A \longrightarrow B$ is said to be a $1-1$ correspondence if $f$ is both $1-1$ and onto.

**Example 3.16**:

1. Let $A=\{0,1,2,3,4,5\}$ and $B=\{0,5,10,15,20,25\}$. Suppose $f: A \to B$ given by $f(x)=5x$ for all $x \in A$. One can easily see that every element of $B$ has a preimage in $A$ and hence $f$ is onto. Moreover, if $f(x)=f(y)$, then $5x=5y$, i.e. $x=y$. Hence, $f$ is $1-1$. Therefore, $f$ is a $1-1$ correspondence between $A$ and $B$.
2. Let $A$ be a finite set. If $f: A \to A$ is onto, then it is one to one.

**Solution**: Let $A=\{a_1,a_2,\cdots,a_n\}$. Then $Range(f)=\{f(a_1), f(a_2),\cdots, f(a_n)\}$. Since $f$ is onto we have $Range(f)=A$.Thus, $A=\{f(a_1), f(a_2),\cdots, f(a_n)\}$, which implies that $f(a_1)$, $f(a_2)$, $\cdots$, $f(a_n)$ are all distinct. Hence, $a_i \neq a_j$ implies $f(a_i) \neq f(a_j)$ for all $1 \leq i, j \leq n$. Therefore, $f$ is $1-1$.

# Inverse of a function

Since a function is a relation , the inverse of a function $f$ is denoted by $f^{-1}$ and is defined by:

$$f^{-1} = \{(y, x) : (x, y) \in f\}$$

For instance, if $f = \{(2,4), (3,6), (1,7)\}$ , then $f^{-1} = \{(4,2), (6,3), (7,1)\}$

Note that the inverse of a function is not always a function.

To see this consider the function $f = \{(2,4), (3,6), (5,4)\}$ .

Then, $f^{-1} = \{(4,2), (6,3), (4,5)\}$ , which is not a function.

As we have seen above not all functions have an inverse, so it is important to determine whether or not a function has an inverse before we try to find the inverse. If the function does not have an inverse, then we need to realize that it does not have an inverse so that we do not waste our time trying to find something that does not exist.

A one to one function is special because only one to one functions have inverse. If a function is one to one, to find the inverse we will follow the steps below:

Steps of finding inverse of a function

1. Interchange $x$ and $y$ in the equation $y = f(x)$
2. Solving the resulting equation for $y$, we will obtaining the inverse function.

Note that the domain of the inverse function is the range of the original function and the range of the inverse function is the domain of the original function.

**Example 3.17**:

1. Given $y = f(x) = x^3$. Find $f^{-1}$ and its domain.

**Solution**: We begin by interchanging $x$ and $y$, and we solve for $y$.

| | |
|---|---|
| $y = x^3$ | Interchange $x$ and $y$ |
| $x = y^3$ | Take the cube root of both sides |
| $\sqrt[3]{x} = y$ | This is the inverse of the function |

Thus, $f^{-1}(x) = \sqrt[3]{x}$. The domain of $f^{-1}$ is the set of all real numbers.

2. Let $y = f(x) = \dfrac{x}{x+2}$. Find $f^{-1}(x)$.

**Solution**: Again we begin by interchanging $x$ and $y$, and then we solve for $y$.

$$y = \frac{x}{x+2} \qquad \text{Interchange } x \text{ and } y$$

$$x = \frac{y}{y+2} \qquad \text{Solving for } y$$

$$x(y+2) = y \Leftrightarrow xy + 2x = y \Leftrightarrow 2x = y(1-x) \Leftrightarrow y = \frac{2x}{1-x}$$

Thus, $f^{-1}(x) = \dfrac{2x}{1-x}$

**Remark**: Even though, in general, we use an exponent of $-1$ to indicate a reciprocal, inverse function notation is an exception to this rule. Please be aware that $f^{-1}(x)$ is not the reciprocal of $f$. That is,

$$f^{-1}(x) \neq \frac{1}{f(x)}$$

If we want to write the reciprocal of the function $f(x)$ by using a negative exponent, we must write

$$\frac{1}{f(x)} = \left[f(x)\right]^{-1}.$$

# Polynomials, zeros of polynomials, rational functions and their graphs

## Polynomial functions

**Definition 3.13:** A function of the form $y = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, a_n \neq 0$ is said to be a polynomial function of degree n. Here each coefficients $a_i$ are assumed to be real numbers and n is non negative integer. Also, $a_n$ is called the leading coefficient of the polynomial.

**Remark**: The domain of a polynomial function is always the set of real numbers.

## Types of polynomials

- A polynomial of degree 1 is called a linear function.
- A polynomial of degree 2 is called quadratic function.
- A polynomial of degree 3 is called a cubic function.

  i.e. $\mathbf{p(x) = a_3 x^3 + a_2 x^2 + a_1 x + a_0, a_3 \neq 0}$

**Example 3.18**: $p(x) = 2x^2 + 1$, $q(x) = \sqrt{3}x^4 + 2x - \pi$ and $f(x) = 2x^3$ are examples of polynomial functions.

# Properties of polynomial functions

1) The graph of a polynomial is a smooth unbroken curve. The word smooth means that the graph does not have any sharp corners as turning points.
2) If P is a polynomial of degree n, then it has at most n zeros (the point at which the graph intersects or touches the x-axis).
3) The graph of a polynomial function of degree n can have at most n-1 turning points.
4) The graph of a polynomial shows, as $|x| \to \infty$, $|y| \to \infty$.

# Zeros of a polynomial

The zeros of a polynomial function provide valuable information that can be helpful in sketching its graph.

One can find the zeros by factorizing the polynomial. However, we have no general method for factorizing polynomials of degree greater than 2.

In this subsection, we turn our attention to methods that will allow us to find zeros of higher degree polynomials.

To do this, we first need to discuss about the division algorithm.

## Division Algorithm

Let $p(x)$ and $d(x)$ be polynomials with $d(x) \neq 0$, and with the degree of $d(x)$ less than or equal to the degree of $p(x)$. Then there are polynomials $q(x)$ and $R(x)$ such that

$\underbrace{p(x)}_{dividend} = \underbrace{d(x)}_{divisor}.\underbrace{q(x)}_{quotient} + \underbrace{R(x)}_{remainder}$, where either $R(x) = 0$ or the degree of $R(x)$ is less than degree of $d(x)$.

**Example 3.19**: Divide $\dfrac{x^4 - 1}{x^2 + 2x}$.

**Solution**: Using long division we have

$$
\begin{array}{r}
x^2 - 2x + 4 \\
x^2 + 2x \overline{)\, x^4 + 0x^3 + 0x^2 + 0x - 1} \\
\underline{-(x^4 + 2x^3)} \\
-2x^3 + 0x^2 - 1 \\
\underline{-(-2x^3 - 4x^2)} \\
4x^2 + 0x - 1 \\
\underline{-(4x^2 + 8x)} \\
-8x - 1
\end{array}
$$

This long division means $\underbrace{x^4-1}_{\text{dividend}}=\underbrace{(x^2+2x)}_{\text{divisor}}.\underbrace{(x^2-2x+4)}_{\text{quotient}}+\underbrace{(-8x-1)}_{\text{remainder}}$.

With the aid of the division algorithm, we can derive two important theorems that will allow us to recognize the zeros of polynomials.

If we apply the division algorithm where the divisor, $d(x)$, is linear (that is of the form $x-r$), we get

$$p(x)=(x-r)q(x)+R$$

Note that since the divisor is of the first degree, the remainder $R$, must be a constant. If we now substitute $x=r$, into this equation, we get

$$P(r)=(r-r)q(r)+R=0\cdot q(r)+R$$

Therefore, $p(r)=R$.

The result we just proved is called the remainder theorem.

**The Remainder Theorem**

When a polynomial $p(x)$ of degree at least 1 is divided by $x-r$, then the remainder is $p(r)$.

**Example 3.20**: The remainder when $P(x)=x^3-x^2+3x-1$ is divided by $x-2$ is $p(2)=9$.

As a consequence of the remainder theorem, if $x-r$ is a factor of $p(x)$, then the remainder must be 0. Conversely, if the remainder is 0, then $x-r$, is a factor of $p(x)$. This is known as the Factor Theorem.

**The Factor Theorem**

$x-r$ is a factor of $p(x)$ if and only if $p(r)=0$.

The next theorem, called location theorem, allows us to verify that a zero exists somewhere within an interval of numbers, and can also be used to zoom in closer on a value.

**Location theorem**

Let $f$ be a polynomial function and $a$ and $b$ be real numbers such that $a<b$. If $f(a)f(b)<0$, then there is at least one zero of $f$ between $a$ and $b$.

- Example:

a) Show that x-2 is the factor of

$$P(x) = x^3 - x^2 + 3x - 10$$

b) show that $p(x) = 4x^3 + 3x - 6$ has at least one root between 0 and 1

The Factor and Remainder theorems establish the intimate relationship between the factors of a polynomial $p(x)$ and its zeros. Recall that a polynomial of degree $n$ can have at most $n$ zeros. Does every polynomial have a zero? Our answer depends on the number system in which we are working. If we restrict ourselves to the set of real number system, then we are already familiar with the fact that the polynomial $p(x)=x^2+1$ has no real zeros. However, this polynomial does have two zeros in the complex number system. (The zeros are $i$ and $-i$).

| **Fundamental Theorem of Algebra** |
| --- |
| If $p(x)$ is a polynomial of degree $n>0$ whose coefficients are complex numbers, then $p(x)$ has at least one zero in the complex number system. |

Note that since all real numbers are complex numbers, a polynomial with real coefficients also satisfies the Fundamental theorem of Algebra. As an immediate consequence of the Fundamental theorem of Algebra, we have

## The Fundamental Theorem of Algebra

**If p(x) is a polynomial of degree n ≥ 1, then p(x) = 0 has exactly n roots, including multiplicities and complex roots.**

**The linear Factorization Theorem**

If $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, where $n \geq 1$ and $a_n \neq 0$, then

$p(x) = a_n (x - r_1)(x - r_2) \cdots (x - r_n)$, where the $r_i$ are complex numbers (possible real and not necessarily distinct).

From the linear factorization theorem, it follows that every polynomial of degree $n \geq 1$ has exactly $n$ zeros in the complex number system, where a root of multiplicity $k$ counted $k$ times.

**Example 3.21**: Express each of the polynomials in the form described by the Linear Factorization Theorem. List each zero and its multiplicity.

a) $p(x) = x^3 - 6x^2 - 16x$

b) $q(x) = 3x^2 - 10x + 8$

c) $f(x) = 2x^4 + 8x^3 + 10x^2$

**Solution**:

a) We may factorize $p(x)$ as follows:

$$p(x) = x^3 - 6x^2 - 16x = x(x^2 - 6x - 16)$$
$$= x(x-8)(x+2)$$
$$= x(x-8)(x-(-2))$$

The zeros of $p(x)$ are 0, 8, and – 2 each of multiplicity one.

b) We may factorize $q(x)$ as follows:

$$q(x) = 3x^2 - 10x + 8 = (3x-4)\ (x-2)$$
$$= 3(x - \frac{4}{3})(x-2)$$

Thus, the zeros of $q(x)$ are $\frac{4}{3}$ and 2, each of multiplicity one.

c) We may factorize $f(x)$ as follows:

$$f(x) = 2x^4 + 8x^3 + 10x^2 = 2x^2(x^2 + 4x + 5)$$
$$= 2x^2(x-(-2+i))(x-(-2-i))$$

Thus, the zeros of f(x) are 0 with multiplicity two and $-2+i$ and $-2-i$ each with multiplicity one.

**Example 3.22**:

1. Find a polynomial $p(x)$ with exactly the following zeros and multiplicity.

| zeros | multiplicity |
|---|---|
| $-1$ | 3 |
| 2 | 4 |
| 5 | 2 |

Are there any other polynomials that give the same roots and multiplicity?

2. Find a polynomial $f(x)$ having the zeros described in part (1) such that $f(1) = 32$.

**Solution**:

Based on the Factor Theorem, we may write the polynomial as:

$$p(x) = (x-(-1))^3 (x-2)^4 (x-5)^2 = (x+1)^3 (x-2)^4 (x-5)^2$$

which gives the required roots and multiplicities.

Any polynomial of the form $kp(x)$, where $k$ is a non -zero constant will give the same roots and multiplicities.

2. Based on part (1), we know that $f(x) = k(x+1)^3(x-2)^4 (x-5)^2$. Since we want $f(x) = 32$, we have

$$f(1) = k(1+1)^3 (1-2)^4 (1-5)^2$$

$$32 = k(8)(1)(16) \quad \Rightarrow k = \tfrac{1}{4}$$

Thus, $f(x) = \tfrac{1}{4}(x+1)^3 (x-2)^4 (x-5)^2$.

**Example 3.22**:

1. Find a polynomial $p(x)$ with exactly the following zeros and multiplicity.

| zeros | multiplicity |
|---|---|
| $-1$ | 3 |
| 2 | 4 |
| 5 | 2 |

Are there any other polynomials that give the same roots and multiplicity?

2. Find a polynomial $f(x)$ having the zeros described in part (1) such that $f(1) = 32$.

Our experience in using the quadratic formula on quadratic equations with real coefficients has shown us that complex roots always appear in conjugate pairs. For example, the roots of $x^2 - 2x + 5 = 0$ are $1 + 2i$ and $1 - 2i$. In fact, this property extends to all polynomial equations with real coefficients.

**Conjugate Roots Theorem**

Let $p(x)$ be a polynomial with real coefficients. If complex number $a + bi$ (where $a$ and $b$ are real numbers) is a zero of $p(x)$, then so is its conjugate $a - bi$.

**Example 3.23**: Let $r(x) = x^4 + 2x^3 - 9x^2 + 26x - 20$. Given that $1 - \sqrt{3}\, i$ is a zero, find the other zero of $r(x)$.

**Solution**: According to the Conjugate Roots Theorem, if $1 - \sqrt{3}\, i$ is a zero, then its conjugate, $1 + \sqrt{3}\, i$ must also be a zero. Therefore, $x - (1 - \sqrt{3}\, i)$ and $x - (1 + \sqrt{3}\, i)$ are both factors of $r(x)$, and so their product must be a factor of $r(x)$. That is, $[x - (1 - \sqrt{3}\, i)]\,[x - (1 + \sqrt{3}\, i)] = x^2 - 2x + 4$ is a factor of $r(x)$. Dividing $r(x)$ by $x^2 - 2x + 4$, we obtain

$$r(x) = (x^2 - 2x + 4)(x^2 + 4x - 5) = (x^2 - 2x + 4)(x + 5)(x - 1).$$

Thus, the zeros of $r(x)$ are $1-\sqrt{3}i$, $1+\sqrt{3}i$, $-5$ and 1.

The theorems we have discussed so far are called existence theorems because they ensure the existence of zeros and linear factors of polynomials. These theorems do not tell us how to find the zeros or the linear factors. The Linear Factorization Theorem guarantees that we can factor a polynomial of degree at least one into linear factors, but it does not tell us how.

We know from experience that if $p(x)$ happens to be a quadratic function, then we may find the zeros of $p(x)=Ax^2+Bx+C$ by using the quadratic formula to obtain the zeros

$$x=\frac{-B\pm\sqrt{B^2-4AC}}{2A}.$$

The rest of this subsection is devoted to developing some special methods for finding the zeros of a polynomial function.

As we have seen, even though we have no general techniques for factorizing polynomials of degree greater than 2, if we happen to know a root, say $r$, we can use long division to divide $p(x)$ by $x-r$ and obtain a quotient polynomial of lower degree. If we can get the quotient polynomial down to a quadratic, then we are able to determine all the roots. But how do we find a root to start the process? The following theorem can be most helpful.

## The Rational Root Theorem

Suppose that $f(x)=a_n x^n+a_{n-1}x^{n-1}+\cdots+a_1x+a_0$, where $n\geq 1$, $a_n\neq 0$ is an $n^{th}$ degree polynomial with integer coefficients. If $\frac{p}{q}$ is a rational root of $f(x)=0$, where $p$ and $q$ have no common factor other than $\pm 1$, then $p$ is a factor of $a_0$ and $q$ is a factor of $a_n$.

To get a feeling as to why this theorem is true, suppose $\frac{3}{2}$ is a root of

$a_3x^3+a_2x^2+a_1x+a_0=0$.

Then, $a_3\left(\frac{3}{2}\right)^3+a_2\left(\frac{3}{2}\right)^2+a_1\left(\frac{3}{2}\right)+a_0=0$ which implies that

$$\frac{27a_3}{8}+\frac{9a_2}{4}+\frac{3a_1}{2}+a_0=0$$ multiplying both sides by 8

$$27a_3+18a_2+12a_1=-8a_0 \quad \text{.....(1)}$$

$$27a_3=-18a_2-12a_1-8a_0 \quad \text{.....(2)}$$

If we look at equation (1), the left hand side is divisible by 3, and therefore the right hand side must also be divisible by 3. Since 8 is not divisible by 3, $a_0$ must be divisible by 3. From equation (2), $a_3$ must be divisible by 2.

**Example 3.24**: Find all the zeros of the function $p(x) = 2x^3 + 3x^2 - 23x - 12$.

**Solution**: According to the Rational Root Theorem, if $\frac{p}{q}$ is a rational root of the given equation then $p$ must be a factor of $-12$ and $q$ must be a factor of 2. Thus, we have

possible values of $p$ : $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$

possible values of $q$ : $\pm 1, \pm 2$

possible rational roots $\frac{p}{q}$ : $\pm 1, \pm \frac{1}{2}, \pm 2, \pm 3, \pm \frac{3}{2}, \pm 4, \pm 6, \pm 12$

We may check these possible roots by substituting the value in $p(x)$. Now $p(1) = -30$ and $p(-1) = 12$. Since $p(1)$ is negative and $p(-1)$ is positive, by **Location theorem,** $p(x)$ has a zero between $-1$ and 1. Since $P\left(-\frac{1}{2}\right) = 0$, then $\left(x + \frac{1}{2}\right)$ is a factor of $p(x)$. Using long division, we obtain

$$p(x) = 2x^3 + 3x^2 - 23x - 12 = (x + \tfrac{1}{2})(2x^2 + 2x - 24)$$
$$= 2(x + \tfrac{1}{2})(x + 4)(x - 3)$$

Therefore, the zeros of $p(x)$ are $-\frac{1}{2}$, $-4$ and 3.

**Example3.24**: Find all the zeros of the function $p(x) = 2x^3 + 3x^2 - 23x - 12$.

# Rational Functions and their Graphs

A rational function is a function of the form $f(x)=\frac{n(x)}{d(x)}$ where both $n(x)$ and $d(x)$ are polynomials and $d(x)\neq 0$.

**Example 3.25**: The functions $f(x)=\frac{3}{x+5}$, $f(x)=\frac{x-1}{x^2-4}$ and $f(x)=\frac{x^5+2x^3-x+1}{x\ +5x}$ are examples of rational function.

Note that the domain of the rational function $f(x)=\frac{n(x)}{d(x)}$ is $\{x : d(x)\neq 0\}$

**Example 3.26**: Find the domain and zeros of the function $f(x)=\frac{3x-5}{x^2-x-12}$.

**Solution**: The values of $x$ for which $x^2-x-12=0$ are excluded from the domain of $f$. Since $x^2-x-12=(x-4)(x+3)$, we have $Dom(f)=\{x : x\neq -3,4\}$. To find the zeros of $f(x)$, we solve the equation

$$\frac{n(x)}{d(x)}=0 \Leftrightarrow n(x)=0 \ \&\ q(x)\neq 0$$

Therefore, to find the zeros of $f(x)$, we solve $3x-5=0$, giving $x=\frac{5}{3}$. Since $\frac{5}{3}$ does not make the denominator zero, it is the only zero of $f(x)$.

The following terms and notations are useful in our next discussion.

Given a number $a$,

- $x$ approaches $a$ from the right means $x$ takes any value near and near to $a$ but $x > a$. This is denoted by: $x \to a^+$ (read: '$x$ approaches $a$ from the right' ).

  For instance, $x \to 1^+$ means $x$ can be 1.001, 1.0001, 1.00001, 1.000001, etc.

- $x$ approaches $a$ from the left means $x$ takes any value near and near to $a$ but $x < a$.

  This is denoted by: $x \to a^-$ (read: '$x$ approaches $a$ from the left' ).

  For instance, $x \to 1^-$ means $x$ can be 0.99, 0.999, 0.9999, 0.9999, etc.

- $x \to \infty$ (read: '$x$ approaches or tends to *infinity*') means the value of $x$ gets indefinitely larger and larger in magnitude (keep increasing without bound). For instance, $x$ can be $10^6$, $10^{10}$, $10^{12}$, etc.

$x \to -\infty$ (read: '$x$ approaches or tends to negative *infinity*') means the value of $x$ is negative and gets indefinitely larger and larger negative in magnitude (keep decreasing without bound). For instance, $x$ can be $-10^6$, $-10^{10}$, $-10^{12}$, etc.

The same meanings apply also for the values of a function $f$ if we wrote $f(x)\to\infty$ or $f(x)\to-\infty$. The following figure illustrates these notion and notations.

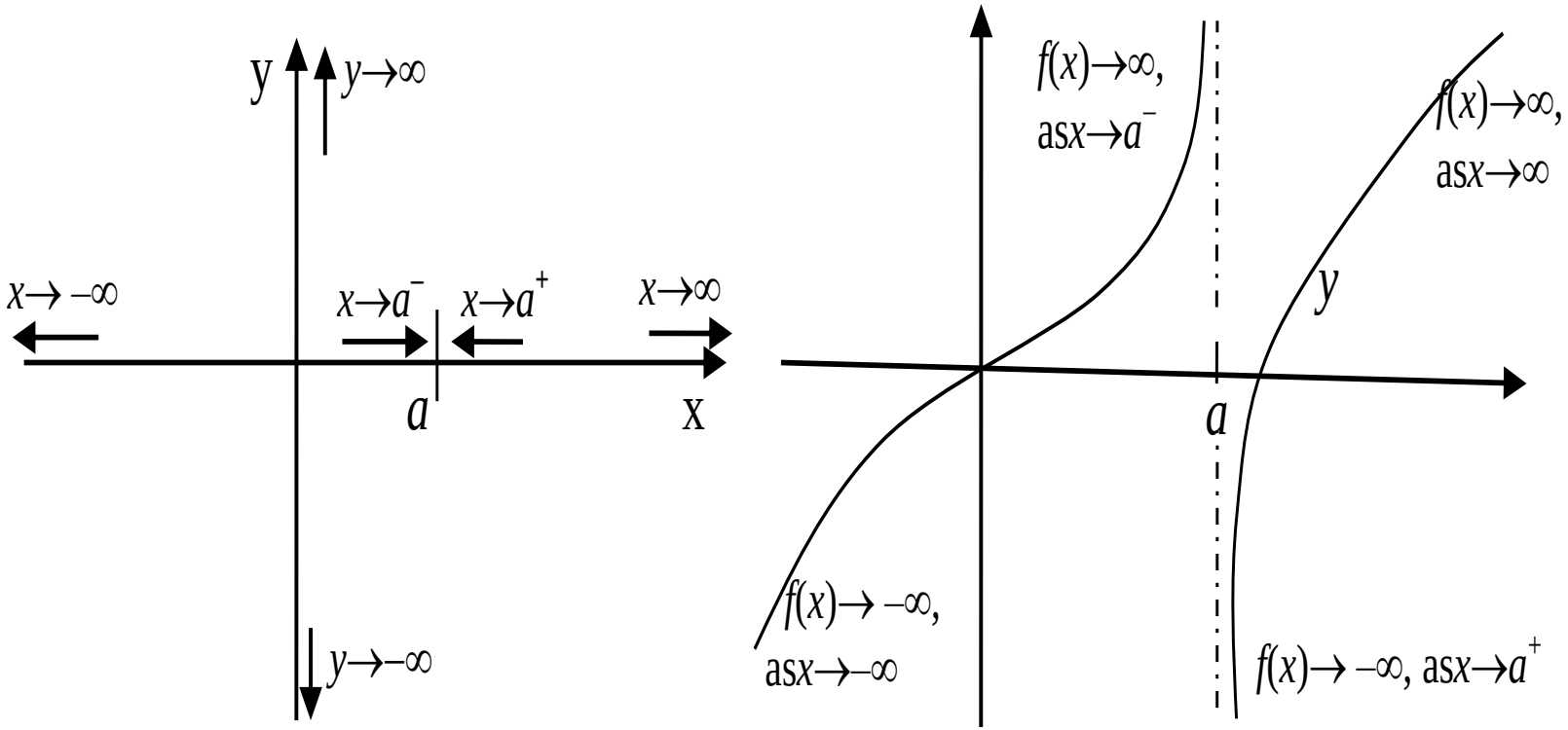



Fig. 2.1. Graphical illustration of the idea of $x\to a^+$, $f(x)\to\infty$, etc.

We may also write $f(x)\to b$ (read: '$f(x)$ approaches $b$') to mean the function values, $f(x)$, becomes arbitrarily closer and closer to $b$ (i.e., approximately $b$) but not exactly equal to $b$. For instance, if $f(x)=\frac{1}{x}$, then $f(x)\to 0$ as $x\to\infty$; i.e., $\frac{1}{x}$ is approximately 0 when $x$ is arbitrarily large.

The following steps are usually used to sketch (or draw) the graph of a rational function $f(x)$.

1. Identify the domain and simplify it.

2. Find the intercepts of the graph whenever possible. Recall the following:
   - y–intercept is the point on y-axis where the graph of $y = f(x)$ intersects with the $y$-axis. At this point $x=0$. Thus, $y = f(0)$, or $(0, f(0))$ is the y-intercept if $0\in \text{Dom}(f)$.
   - x–intercept is the point on x-axis where the graph of $y = f(x)$ intersects with the $x$-axis. At this point $y=0$. Thus, $x=a$ or $(a, 0)$ is x-intercept if $f(a)=0$.

3. Determine the asymptotes of the graph. Here, remember the following.

- Vertical Asymptote: The vertical line $x=a$ is called a vertical asymptote(VA) of $f(x)$ if
  i) $a\notin \text{dom}(f)$, i.e., $f$ is not defined at $x=a$; and
  ii) $f(x)\to\infty$ or $f(x)\to -\infty$ when $x\to a^+$ or $x\to a^-$. In this case, the graph of $f$ is almost vertically rising upward (if $f(x)\to\infty$) or sinking downward (if $f(x)\to-\infty$) along with the vertical line $x=a$ when $x$ approaches $a$ either from the right or from the left.

**Example 3.27**: Consider $f(x)=\frac{1}{(x-a)^n}$, where $a \geq 0$ and $n$ is a positive integer.

Obviously $a \notin \text{Dom}(f)$. Next, we investigate the trend of the values of $f(x)$ near $a$. To do this, we consider two cases, when $n$ is even or odd:

Suppose $n$ is even: In this case $(x-a)^n > 0$ for all $x \in \Re \setminus \{a\}$; and since $(x-a)^n \to 0$ as $x \to a^+$ or $x \to a^-$. Hence, $f(x)=\frac{1}{(x-a)^n} \to \infty$ as $x \to a^+$ or $x \to a^-$. Therefore, $x=a$ is a VA of $f(x)$. Moreover, $y= 1/a^n$ or $(0, 1/a^n)$ is its y-intercept since $f(0)=1/a^n$. However, it has no x-intercept since $f(x) >0$ for all $x$ in its domain (See, Fig. 2.2 (A)).

Suppose $n$ is odd: In this case $(x-a)^n > 0$ for all $x>a$ and $1/(x-a)^n \to \infty$ when $x \to a^+$ as in the above case. Thus, $x=a$ is its VA. However, $1/(x-a)^n \to -\infty$ when $x \to a^-$ since $(x-a)^n < 0$ for $x<a$. Moreover, $y= -1/a^n$ or $(0, -1/a^n)$ is its y-intercept since $f(0) = -1/a^n$. However, it has no x-intercept also in this case. (See, Fig. 2.2 (B)).

Note that in both cases, $f(x)=\frac{1}{(x-a)^n} \to 0$ as $x \to \infty$ or $x \to -\infty$.

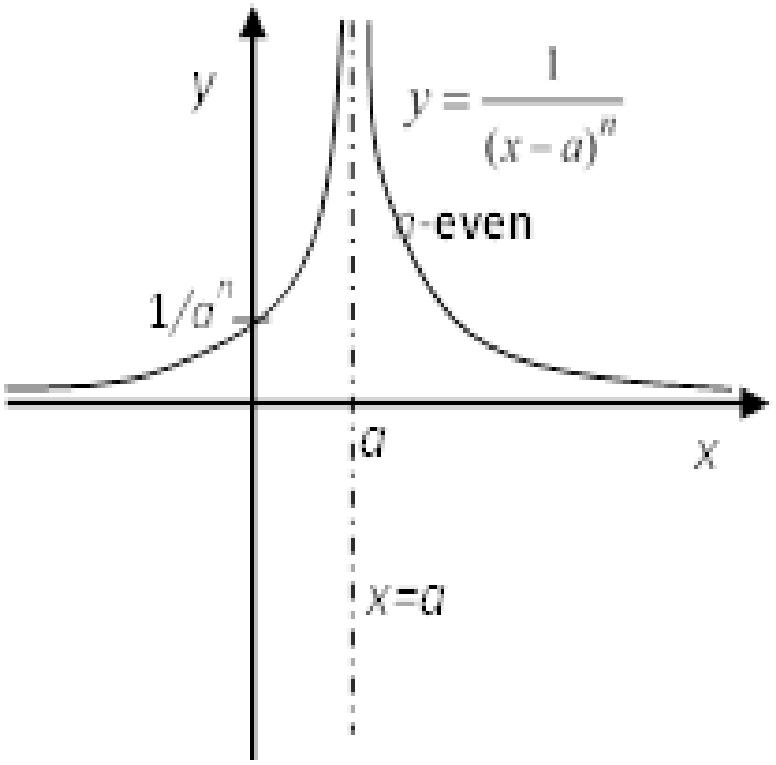



Fig. 2.2 (A)

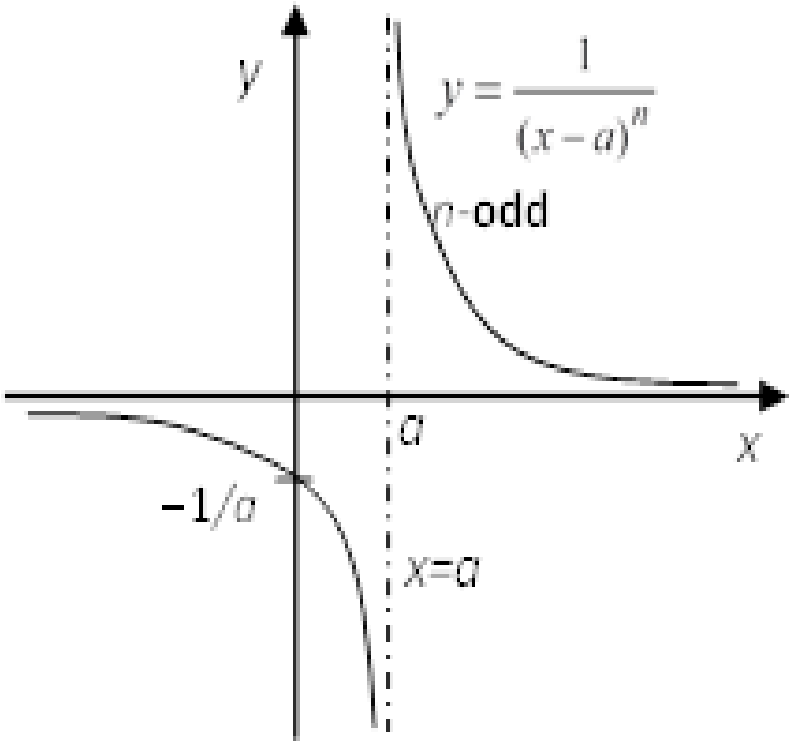



Fig. 2.2 (B)

**Remark**: Let $f(x)=\dfrac{n(x)}{d(x)}$ be a rational function. Then,

1. if $d(a)=0$ and $n(a)\neq 0$, then $x=a$ is a VA of $f$ .
2. if $d(a)=0=n(a)$, then $x=a$ may or may **not** be a VA of $f$. In this case, simplify $f(x)$ and look for VA of the simplest form of $f$.

- Horizontal Asymptote: A horizontal line $y=b$ is called horizontal asymptote (HA) of $f(x)$ if the value of the function becomes closer and closer to $b$ (i.e., $f(x)\to b$)as $x\to\infty$ or as $x\to -\infty$.

  In this case, the graph of $f$ becomes almost a horizontal line along with (or near) the line $y=b$ as $x\to\infty$ and as $x\to-\infty$. For instance, from the above example, the HA of $f(x)=\frac{1}{(x-a)^n}$ is $y=0$ (the x-axis) , for any positive integer $n$ (See, Fig. 2.2).

**Remark**: A rational function $f(x)=\frac{n(x)}{d(x)}$ has a HA only when $degree(n(x)) \le degree(d(x))$.

In this case, (i) If $degree(n(x)) < degree(d(x))$, then $y = 0$ (the x-axis) is the HA of $f$.

(ii) If $degree(n(x)) = degree(d(x)) = n$, i.e., $f(x)=\frac{a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0}{b_n x^n + b_{n-1}x^{n-1} + \cdots + b_1 x + b_0}$,

then $y=\frac{a_n}{b_n}$ is the HA of $f$.

- Oblique Asymptote: The oblique line $y=ax+b$, $a\neq 0$, is called an oblique asymptote (OA) of $f$ if the value of the function, $f(x)$, becomes closer and closer to $ax+b$(i.e., $f(x)$ becomes approximately $ax+b$) as either $x\to\infty$ or $x\to -\infty$. In this case, the graph of $f$ becomes almost a straight line along with (or near) the oblique line $y=ax+b$ as $x\to\infty$ and as $x\to -\infty$.

**Note**: A rational function $f(x)=\frac{n(x)}{d(x)}$ has an OA only when *degree*($n(x)$) = *degree*($d(x)$) + 1. In this case, using long division, if the quotient of $n(x) \div d(x)$ is $ax+b$, then $y=ax+b$ is the OA of $f$.

**Example 3.28:** Sketch the graphs of (a) $f(x)=\frac{x+2}{x-1}$ (b) $g(x)=\frac{x^2+3x+2}{x^2-1}$

**Solution**: (a) Since $x-1=0$ at $x=1$, dom($f$) = $\Re\backslash\{1\}$.

- Intercepts: y-intercept: $x=0 \Rightarrow y=f(0)=-2$. Hence, $(0,-2)$ is y-intercept.
  x-intercept: $y=0 \Rightarrow x+2=0 \Rightarrow x=-2$. Hence, $(-2,0)$ is x-intercept.
- Asymptotes:
  - VA: Since $x-1=0$ at $x=1$ and $x+2\neq 0$ at $x=1$, $x=1$ is VA of $f$. In fact, if $x\to 1^+$, then $x+2\approx 3$ but the denominator $x-1$ is almost 0 (but positive).
    Consequently, $f(x)\to\infty$ as $x\to 1^+$.
    Moreover, $f(x)\to -\infty$ as $x\to 1^-$ (since , if $x\to 1^-$ then $x-1$ is almost 0 but negative ) .

    (So, the graph of $f$ rises up to $+\infty$ at the right side of $x=1$, and sink down to $-\infty$ at the left side of $x=1$)
  - HA: Note that if you divide $x+2$ by $x-1$, the quotient is 1 and remainder is 3. Thus,
    $f(x)=\frac{x+2}{x-1}=1+\frac{3}{x-1}$. Thus, if $x\to\infty$ (or $x\to -\infty$), then $\frac{3}{x-1}\to 0$ so that $f(x)\to 1$.
    Hence, $y=1$ is the HA of $f$.

**Example 3.28:** Sketch the graphs of (a) $f(x)=\dfrac{x+2}{x-1}$ (b) $g(x)=\dfrac{x^2+3x+2}{x^2-1}$

Using these information, you can sketch the graph of $f$ as displayed below in Fig. 2.3 (A).

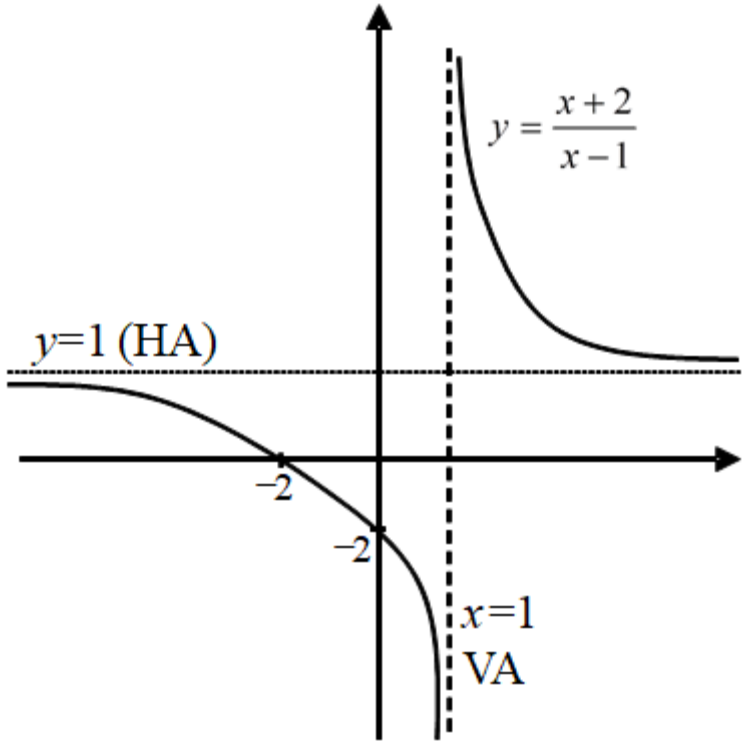



Fig 2.3 (A) $f(x) = \frac{x+2}{x-1}$

(b) The denominator is 0 at $x=1$. So, first factorize and simplify them:

$$x^2+3x+2=(x+2)(x+1) \text{ and } x^2-1=(x-1)(x+1).$$

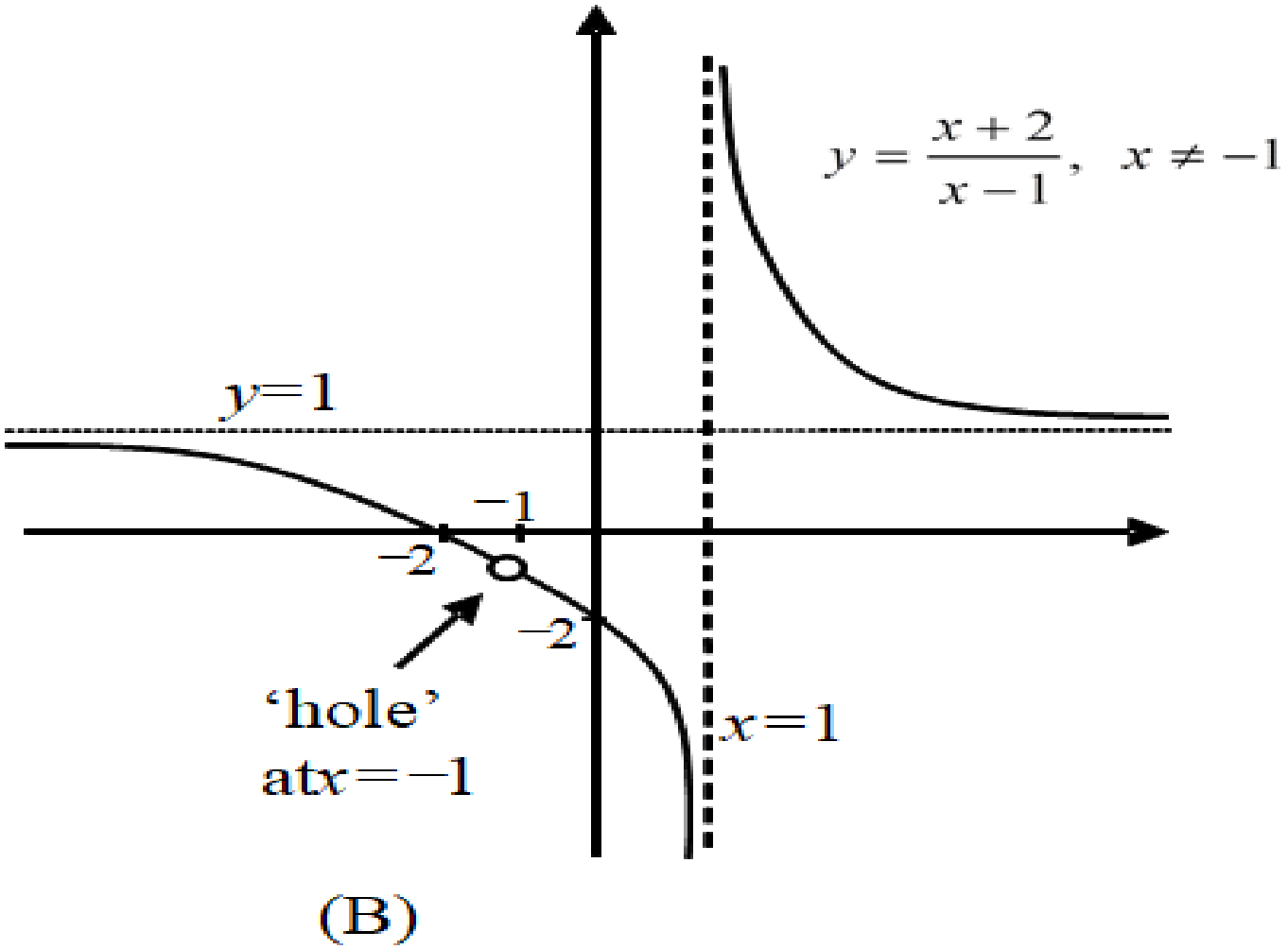
Therefore, $g(x)=\dfrac{x^2+3x+2}{x^2-1}=\dfrac{(x+2)(x+1)}{(x-1)(x+1)}, \quad x\neq -1$

$$=\frac{x+2}{x-1}. \qquad (\text{So, } \mathrm{Dom}(g)=\Re\setminus\{1,-1\})$$

This implies that only $x=1$ is VA.

Hence, the graph of $g(x)=\dfrac{x+2}{x-1}, \quad x\neq -1,$

is exactly the same as that of $f(x)=\dfrac{x+2}{x-1}$ except that $g(x)$ is not defined at $x=-1$. Therefore, the graph of $g$ and its VA are the same as that of $f$ except that there should be a 'hole' at the point corresponding to $x=-1$ on the graph of $g$ as shown on Fig. 2.3(B) below.




(B)

# Exercise

Find the domain and the real zeros of the given functions and sketch their graphs.

(a) $f(x) = \dfrac{3x}{x^2+2x-3}$.

(b) $f(x) = \dfrac{x^3+2x^2-x-2}{x^2+6x+8}$.

(c) $f(x) = \dfrac{(x+2)(x-3)}{(x+1)^2(x-2)}$.

(d) $f(x) = \dfrac{(x+2)^2(x-2)}{2(x-1)^2(x-3)}$.

# Definition and basic properties of logarithmic, exponential, trigonometric and hyperbolic functions and their graphs

- **Exponents and radicals**

**Definition 3.14:** For a natural number $n$ and a real number $x$,

the power $x^n$, read "the $n^{th}$ power of $x$" or "$x$ raised to $n$", is defined as:

$$x^n = \underbrace{x \cdot x \cdot \cdots \cdot x}_{n \text{ factors each equal to } x}$$

In the symbol $x^n$, $x$ is called the base and $n$ is called the exponent.

**Definition 3.15: (Zero and Negative Exponents)**

*Definition of zero Exponent*

$$x^0 = 1 \ (x \neq 0)$$

*Definition of Negative Exponent*

$$x^{-n} = \frac{1}{x^n} \quad (x \neq 0)$$

Note: $0^0$ is undefined.

As a result of the above definition, we have $\frac{1}{x^{-n}} = x^n$.

We have the following rules of exponents for integer exponents:

**Rules for Integer Exponents**

1. $x^n \cdot x^m = x^{n+m}$
2. $(x^n)^m = x^{nm}$
3. $\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n} \quad (y \neq 0)$
4. $(xy)^n = x^n y^n$
5. $\frac{x^n}{x^m} = x^{n-m}$

Next we extend the definition of exponents even further to include rational number exponents. To do this, we assume that we want the rules for integer exponents also to apply to rational exponents and then use the rules to show us to define a rational exponent. For example, how do we define $a^{\frac{1}{2}}$? Consider $9^{\frac{1}{2}}$.

If we apply rule 2 and square $9^{\frac{1}{2}}$, we get $\left(9^{\frac{1}{2}}\right)^2 = 9^{\frac{2}{2}} = 9$. Thus, $9^{\frac{1}{2}}$ is a number that, when squared, yields 9. There are two possible answers: 3 and – 3, since squaring either number will yield 9. To avoid ambiguity, we define $a^{\frac{1}{2}}$ (called the principal square root of $a$) as the non-negative quantity that, when squared, yield $a$. Thus, $9^{\frac{1}{2}} = 3$.

We will arrive at the definition of $a^{\frac{1}{3}}$ in the same way as we did for $a^{\frac{1}{2}}$. For example, if we cube $8^{\frac{1}{3}}$, we get $\left(8^{\frac{1}{3}}\right)^3 = 8^{\frac{3}{3}} = 8$. Thus, $8^{\frac{1}{3}}$ is the number that, when cubed, yields 8. Since $2^3 = 8$ we have $8^{\frac{1}{3}} = 2$. Similarly, $\left(-27\right)^{\frac{1}{3}} = -3$. Thus, we define $a^{\frac{1}{3}}$ (called the cube root of $a$) as the quantity that, when cubed yields $a$.

**Definition 3.16: (Rational Exponent $a^{\frac{1}{n}}$)**

If $n$ is an odd positive integer, then $a^{\frac{1}{n}} = b$ if and only if $b^n = a$

If $n$ is an even positive integer and $a \geq 0$, then $a^{\frac{1}{n}} = |b|$ if and only if $b^n = a$

We call $a^{\frac{1}{n}}$ the principal $n^{th}$ root of $a$. Hence, $a^{\frac{1}{n}}$ is the real number (nonnegative when $n$ is even) that, when raised to the $n^{th}$ power, yields $a$. Therefore,

$(16)^{\frac{1}{2}} = 4$ since $4^2 = 16$

$(-125)^{\frac{1}{3}} = -5$ since $(-5)^3 = -125$

$\left(\frac{1}{81}\right)^{\frac{1}{4}} = \frac{1}{3}$ since $\left(\frac{1}{3}\right)^4 = \frac{1}{81}$

$27^{\frac{1}{3}} = 3$ since $3^3 = 27$

$(-16)^{\frac{1}{4}}$ is not a real number

Thus far, we have defined $a^{\frac{1}{n}}$, where $n$ is a natural number. With the help of the second rule for exponent, we can define the expression $a^{\frac{m}{n}}$, where $m$ and $n$ are natural numbers and $\frac{m}{n}$ is reduced to lowest terms.

**Definition 3.17: (Rational Exponent $a^{\frac{m}{n}}$)**

If $a^{\frac{1}{n}}$ is a real number, then $a^{\frac{m}{n}} = \left(a^{\frac{1}{n}}\right)^m$ (i.e. the $n^{th}$ root of a raised to the $m^{th}$ power)

We can also define negative rational exponents:

$$a^{-\frac{m}{n}} = \frac{1}{a^{\frac{m}{n}}} \quad (a \neq 0)$$

**Example 3.29**: Evaluate the following

a) $27^{\frac{2}{3}}$ b) $36^{-\frac{1}{2}}$ c) $(-32)^{-\frac{3}{5}}$

**Solution**: We have

a) $27^{\frac{2}{3}} = \left(27^{\frac{1}{3}}\right)^2 = 3^2 = 9$

b) $36^{-\frac{1}{2}} = \dfrac{1}{36^{\frac{1}{2}}} = \dfrac{1}{6}$

c) $(-32)^{-\frac{3}{5}} = \dfrac{1}{(-32)^{\frac{3}{5}}} = \dfrac{1}{\left((-32)^{\frac{1}{5}}\right)^3} = \dfrac{1}{(-2)^3} = -\dfrac{1}{8}$

Radical notation is an alternative way of writing an expression with rational exponents. We define for real number $a$, the $n^{th}$ root of $a$ as follows:

**Definition 3.18 ($n^{th}$ root of $a$):** $\sqrt[n]{a} = a^{\frac{1}{n}}$, where $n$ is a positive integer.

The number $\sqrt[n]{a}$ is also called the principal $n^{th}$ root of $a$. If the $n^{th}$ root of $a$ exists, we have:

For $a$ a real number and $n$ a positive integer,

$$\sqrt[n]{a^n} = \begin{cases} |a|, & \text{if } n \text{ is even} \\ a, & \text{if } n \text{ is odd} \end{cases}$$

For example, $\sqrt[3]{5^3} = 5$ and $\sqrt[4]{(-3)^4} = 3$.

- **Exponential Functions**

In the previous sections we examined functions of the form $f(x) = x^n$, where $n$ is a constant. How is this function different from $f(x) = n^x$.

**Definition 3.19:** A function of the form $y = f(x) = b^x$, where $b > 0$ and $b \neq 1$, is called an exponential function.

**Example 3.30**: The functions $f(x) = 2^x$, $g(x) = 3^x$ and $h(x) = \left(\frac{1}{2}\right)^x$ are examples of exponential functions.

As usual the first question raised when we encounter a new function is its domain. Since rational exponents are well defined, we know that any rational number will be in the domain of an exponential function. For example, let $f(x) = 3^x$. Then as $x$ takes on the rational values $x = 4$, $-2$, $\frac{1}{2}$ and $\frac{4}{5}$, we have

$$f(4) = 3^4 = 3 \cdot 3 \cdot 3 \cdot 3 = 81 \qquad f(-2) = 3^{-2} = \frac{1}{3^2} = \frac{1}{9}$$

$$f\left(\tfrac{1}{2}\right) = 3^{\frac{1}{2}} = \sqrt{3} \qquad f\left(\tfrac{4}{5}\right) = 3^{\frac{4}{5}} = \sqrt[5]{3^4} = \sqrt[5]{81}$$

Note that even though we do not know the exact values of $\sqrt{3}$ and $\sqrt[5]{81}$, we do know exactly what they mean. However, what about $f(x)$ for irrational values of $x$? For instance, $f(\sqrt{2}) = 3^{\sqrt{2}} = ?$

We have not defined the meaning of irrational exponents. In fact, a precise formal definition of $b^x$ where $x$ is irrational requires the ideas of calculus. However, we can get an idea of what $3^{\sqrt{2}}$ should be by using successive rational approximations to $\sqrt{2}$. For example, we have

$$1.414 < \sqrt{2} < 1.415$$

Thus, it would seem reasonable to expect that $3^{1.414} < 3^{\sqrt{2}} < 3^{1.415}$. Since 1.414 and 1.415 are rational numbers, $3^{1.414}$ and $3^{1.415}$ are well defined, even though we cannot compute their values by hand. Using a calculator, we get $4.7276950 < 3^{\sqrt{2}} < 4.7328918$. If we use better approximations to $\sqrt{2}$, we get $3^{1.4142} < 3^{\sqrt{2}} < 3^{1.4143}$. Using a calculator again, we get $4.7287339 < 3^{\sqrt{2}} < 4.7292535$. Computing $3^{\sqrt{2}}$ directly on a calculator gives $3^{\sqrt{2}} \approx 4.7288044$. This numerical evidence suggests that as $x$ approaches $\sqrt{2}$, the values of $3^x$ approach a unique real number that we designate by $3^{\sqrt{2}}$, and so we will accept without proof, the fact that the domain of the exponential function is the set of real numbers.

The exponential function $y = b^x$, where $b > 0$ and $b \neq 1$, is defined for all real values of $x$. In addition all the rules for rational exponents hold for real number exponents as well.

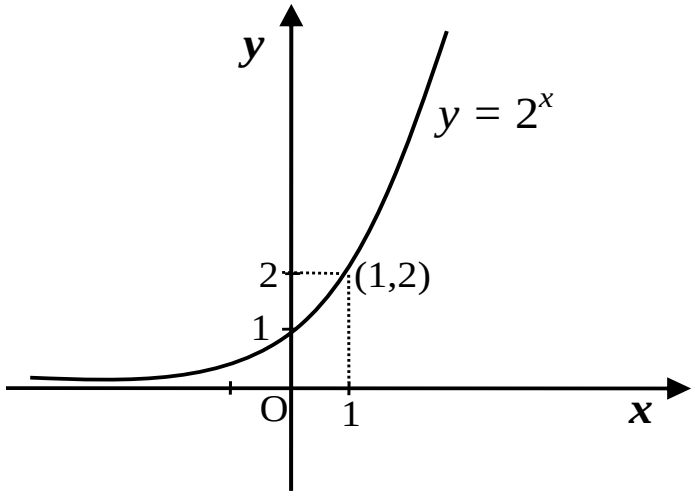
Before we state some general facts about exponential functions , let's see if we can determine what the graph of an exponential function will look like.

**Example 3.31**:

1. Sketch the graph of the function $y = 2^x$ and identify its domain and range.

   **Solution**: To aid in our analysis, we set up a short table of values to give us a frame of reference.

| $x$ | $y$ |
|---|---|
| $-3$ | $2^{-3} = \frac{1}{8}$ |
| $-2$ | $2^{-2} = \frac{1}{4}$ |
| $-1$ | $2^{-1} = \frac{1}{2}$ |
| $0$ | $2^0 = 1$ |
| $1$ | $2^1 = 2$ |
| $2$ | $2^2 = 4$ |
| $3$ | $2^3 = 8$ |

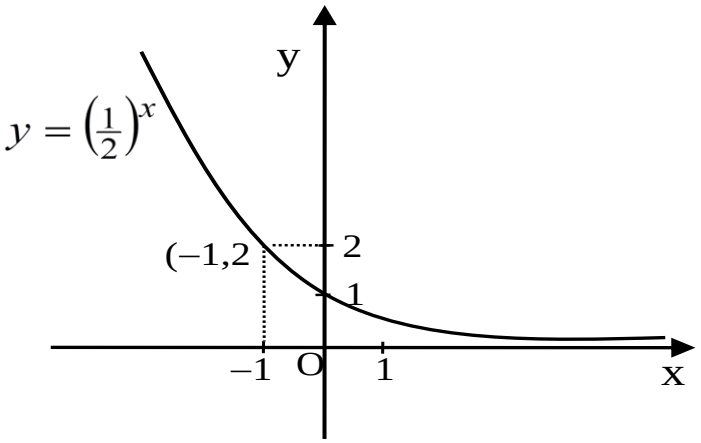


With these points in hand, we draw a smooth curve through the points obtaining the graph appearing above. Observe that the domain of $y = 2^x$ is $IR$, the graph has no $x-$intercepts, as $x \to +\infty$, the $y$ values are increasing very rapidly, whereas as $x \to -\infty$, the $y$ values are getting closer and closer to 0. Thus, $x-axis$ is a horizontal asymptote, the $y-$intercept is 1 and the range of $y = 2^x$ is the set of positive real numbers.

(0,1)

1. Sketch the graph of $y = f(x) = \left(\frac{1}{2}\right)^x$.

**Solution**: It would be instructive to compute a table of values as we did in example 1 above (you are urged to do so). However, we will take a different approach. We note that $y = f(x) = \left(\frac{1}{2}\right)^x = \frac{1}{2^x} = 2^{-x}$. If $f(x) = 2^x$, then $f(-x) = 2^{-x}$. Thus by the graphing principle for $f(-x)$, we can obtain the graph of $y = 2^{-x}$ by reflecting the graph of $y = 2^x$ about the $y-axis$.




Here again the $x-axis$ is a horizontal asymptote, there is no $x-$intercept, 1 is $y-$intercept and the range is the set of positive real numbers. However, the graph is now decreasing rather than increasing.

The following box summarizes the important facts about exponential functions and their graphs.

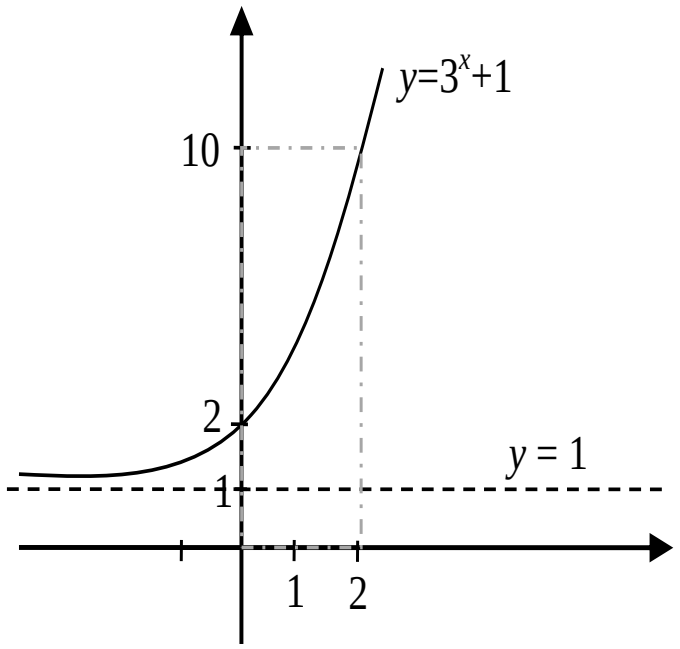
| **The Exponential function** $y=f(x)=b^x$ |
|---|
| 1. The domain of the exponential function is the set of real numbers<br>2. The range of the exponential function is the set of positive real numbers<br>3. The graph of $y=b^x$ exhibits exponential growth if $b>1$ or exponential decay if $0<b<1$.<br>4. The $y-$intercept is $(0,1)$<br>5. The $x-$ **axis** is a horizontal asymptote<br>6. The exponential function is 1 – 1. Algebraically if $b^x=b^y$, then $x=y$ |

**Example 3.32**: Sketch the graph of each of the following. Find the domain, range, intercepts, and asymptotes.

a) $y=3^x+1$ b) $y=3^{x+1}$ c) $y=-9^{-x}+3$

**Solution**:

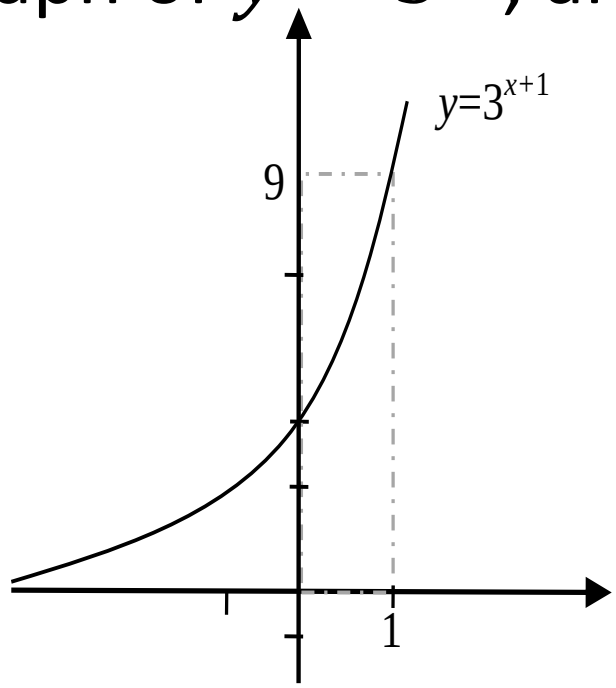
a) To get the graph of $y = 3^x + 1$. We start with the graph of $y = 3^x$, which is the basic exponential growth graph, and shift it up 1 unit.




From the graph we see that

- $Dom(f) = \Re$
- $Range(f) = (1, \infty)$
- The $y-$intercept is $(0,2)$
- The line $y = 1$ is a horizontal asymptote

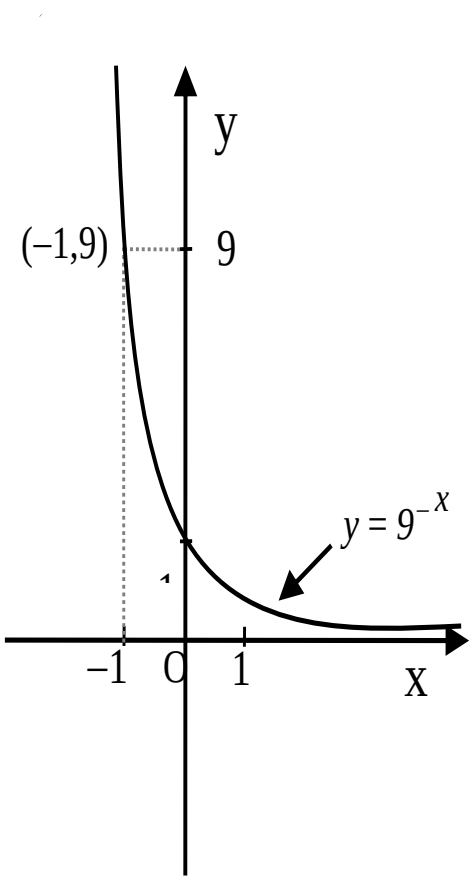
b) To get the graph of $y = 3^{x+1}$ , we start with the graph of $y = 3^{x}$ , and shift $1$ unit to the left.

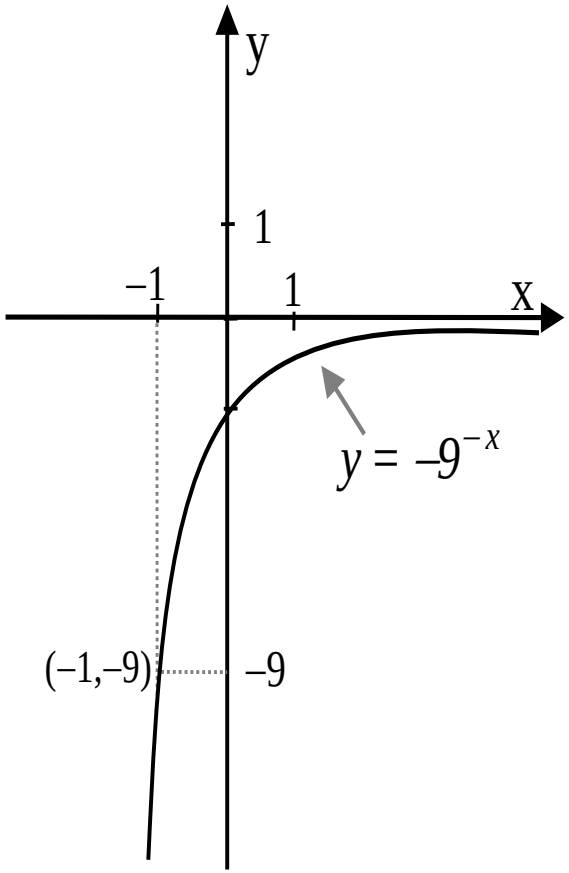


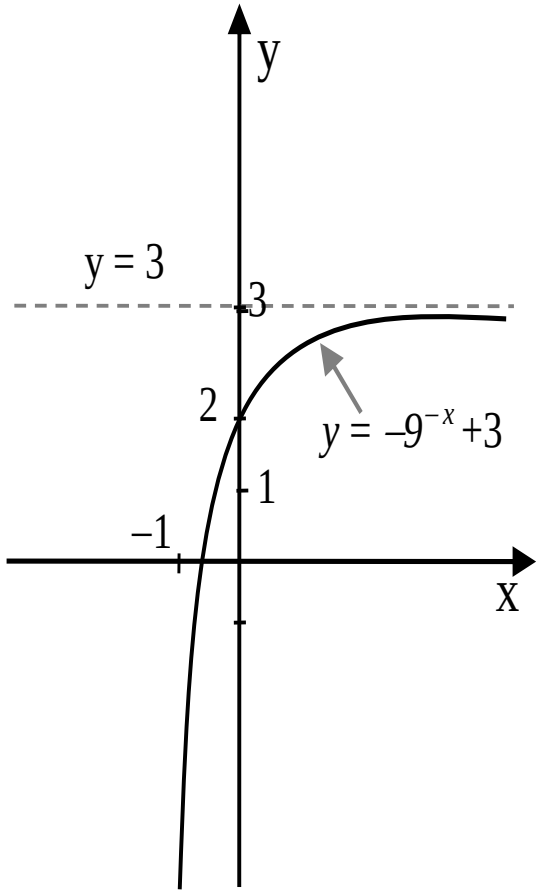

From the graph we see that

- $Dom(f) = \Re$
- $Range(f) = (0, \infty)$
- The $y-$intercept is $(0,3)$
- The line $y = 0$ is a horizontal asymptote

**c) To get the graph of $y = -9^{-x} + 3$, we start with the basic exponential decay $y = 9^{-x}$. We then reflect it with respect to the x-axis, which gives the graph of $y = -9^{-x}$. Finally, we shift this graph up 3 units to get the required graph of $y = -9^{-x} + 3$ .**




From the graph of $y = -9^{-x} + 3$, we can see that $Dom(h) = \Re$, $Range(h) = (-\infty, 3)$, the line $y = 3$ is a horizontal asymptote, 2 is the $y-$intercept and $x = -\frac{1}{2}$ is the $x-$intercept.

**Remark**: When the base $b$ of the exponential function $f(x) = b^x$ equals to the number $e$, where $e = 2.7182\cdots$, we call the exponential function the natural exponential function.

- **Logarithmic Functions**

In the previous subsection we noted that the exponential function $f(x)=b^x$ (where $b>0$ and $b\neq 1$) is one to one. Thus, the exponential function has an inverse function. What is the inverse of $f(x)=b^x$?

To find the inverse of $f(x)=b^x$, let's review the process for finding an inverse function by comparing the process for the polynomial function $y=x^3$ and the exponential function $y=3^x$. Keep in mind that $x$ is our independent variable and $y$ is the dependent variable and so whenever possible we want a function solved explicitly for $y$.

| **To find the inverse of** $y=x^3$ | | **To find the inverse of** $y=3^x$ | |
|---|---|---|---|
| $y=x^3$ | Interchange $x$ and $y$ | $y=3^x$ | Interchange $x$ and $y$ |
| $x=y^3$ | solve for $y$ | $x=3^y$ | solve for $y$ |
| $y=\sqrt[3]{x}$ | | $y=??$ | |

There is no algebraic procedure we can use to solve $x = 3^y$ for $y$. By introducing radical notations we could express the inverse of $y = x^3$ explicitly in the form $y = \sqrt[3]{x}$. In words, $y^3 = x$ and $y = \sqrt[3]{x}$ both mean exactly the same thing: $y$ is the number whose cube is $x$. Similarly, if we want to express $x = 3^y$ explicitly as a function of $x$, we need to invent a special notation for this. The key idea is to take the equation $x = 3^y$ and express it verbally.

$x = 3^y$ means $y$ is the exponent to which 3 must be raised to yield $x$

We introduce the following notation, which expresses this idea in a much more compact form.

**Definition 3.20:** For $b > 0$ and $b \neq 1$, we write $y = \log_b x$ to mean $y$ is the exponent to which $b$ must be raised to yield $x$. In other words,

$$x = b^y \Leftrightarrow y = \log_b x$$

We read $y = \log_b x$ as "$y$ equals the logarithm of $x$ to the base $b$".

REMEMBER: $y = \log_b x$ is an alternative way of writing $x = b^y$

When an expression is written in the form $x = b^y$, it is said to be in exponential form. When an expression is written in the form $y = \log_b x$, it is said to be in logarithmic form. The table below illustrates the equivalence of the exponential and logarithmic forms.

| Exponential form | Logarithmic form |
|---|---|
| $4^2 = 16$ | $\log_4 16 = 2$ |
| $2^4 = 16$ | $\log_2 16 = 4$ |
| $5^{-3} = \frac{1}{125}$ | $\log_5 \frac{1}{125} = -3$ |
| $6^{\frac{1}{2}} = \sqrt{6}$ | $\log_6 \sqrt{6} = \frac{1}{2}$ |
| $7^0 = 1$ | $\log_7 1 = 0$ |

# Logarithmic Function

exponent

$$\log_a x = y \quad \text{means} \quad a^y = x$$

base

$$a > 0,\ a \neq 1, y \neq 0$$

*Example:*

$$\log_2 8 = 3 \quad \text{means} \quad 2^3 = 8$$

**Example 3.33**:

1. Write each of the following in exponential form.

   a) $\log_{3^{\frac{1}{9}}} = -2$ b) $\log_{16} 2 = \frac{1}{4}$

   **Solution**: We have a) $\log_{3^{\frac{1}{9}}} = -2$ means $3^{-2} = \frac{1}{9}$ and b) $\log_{16} 2 = \frac{1}{4}$ means $16^{\frac{1}{4}} = 2$

2. Write each of the following in logarithmic form.

   a) $10^{-3} = 0.001$ b) $27^{\frac{2}{3}} = 9$

   **Solution**: We have a) $10^{-3} = 0.001$ means $\log_{10} 0.001 = -3$

   b) $27^{\frac{2}{3}} = 9$ means $\log_{27} 9 = \frac{2}{3}$

3. Evaluate each of the following.

   a) $\log_3 81$ b) $\log_8 \frac{1}{64}$

**Solution**:

a) To evaluate $\log_3 81$, we let $t = \log_3 81$, and then rewrite the equation in exponential form, $3^t = 81$. Now, if we can express both sides in terms of the same base, we can solve the resulting exponential equation, as follows:

| | | |
|---|---|---|
| Let | $t = \log_3 81$ | Rewrite in exponential form |
| | $3^t = 81$ | Express both sides in terms of the same base |
| | $3^t = 3^4$ | Since the exponential function is 1 – 1 |
| | $t = 4$ | |

Therefore, $\log_3 81 = 4$.

b) We apply the same procedure as in part (a).

| | | |
|---|---|---|
| Let | $t = \log_8 \frac{1}{64}$ | Rewrite in exponential form |
| | $8^t = \frac{1}{64}$ | Express both sides in terms of the same base |
| | $8^t = 8^{-2}$ | Since the exponential function is 1 – 1 |
| | $t = -2$ | |

Therefore, $\log_8 \frac{1}{64} = -2$.

As was pointed out at the beginning of this subsection, logarithm notation was invented to express the inverse of the exponential function. Thus, $\log_b x$ is a function of $x$. We usually write $f(x)=\log_b x$ rather than writing $f(x)=\log_b(x)$ and use parenthesis only when needed to clarify the input to the log function. For example,
If $f(x)=\log_5(4-x)$, then $f(-1)=\log_5(4-(-1))=\log_5 5=1$, whereas if $f(x)=4-\log_5 x$, then $f(-1)=4-\log_5(-1)$, which is undefined.

**Example 3.34**: Given $f(x)=\log_5 x$ , find

a) $f(25)$ b) $f(\frac{1}{25})$ c) $f(0)$ d) $f(-125)$

**Solution**:

a) $f(25)=\log_5 25=2$ (since $5^2=25$)

b) $f(\frac{1}{25})=\log_5 \frac{1}{25}=-2$ (since $5^{-2}=\frac{1}{25}$)

c) $f(0)=\log_5 0$ is not defined (what power of 5 will yield 0?). We say that 0 is not in the domain of $f$.

d) $f(-125)=\log_5(-125)$ is not defined (what power of 5 will yield -125?). We say that -125 is not in the domain of $f$.

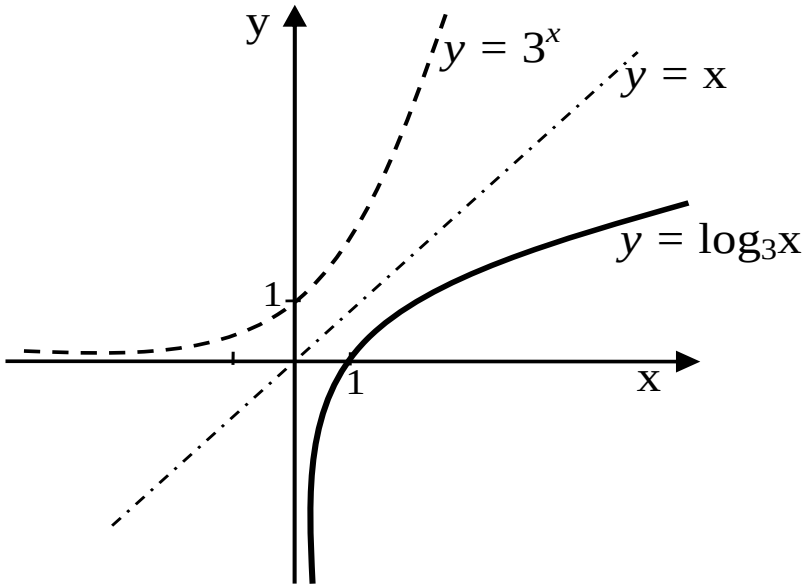
The graph of the inverse function is obtained by reflecting the original graph across the line y = x.

Acknowledging that the logarithmic and exponential functions are inverses, we can derive a great deal of information about the logarithmic function and its graph from the exponential function and its graph.

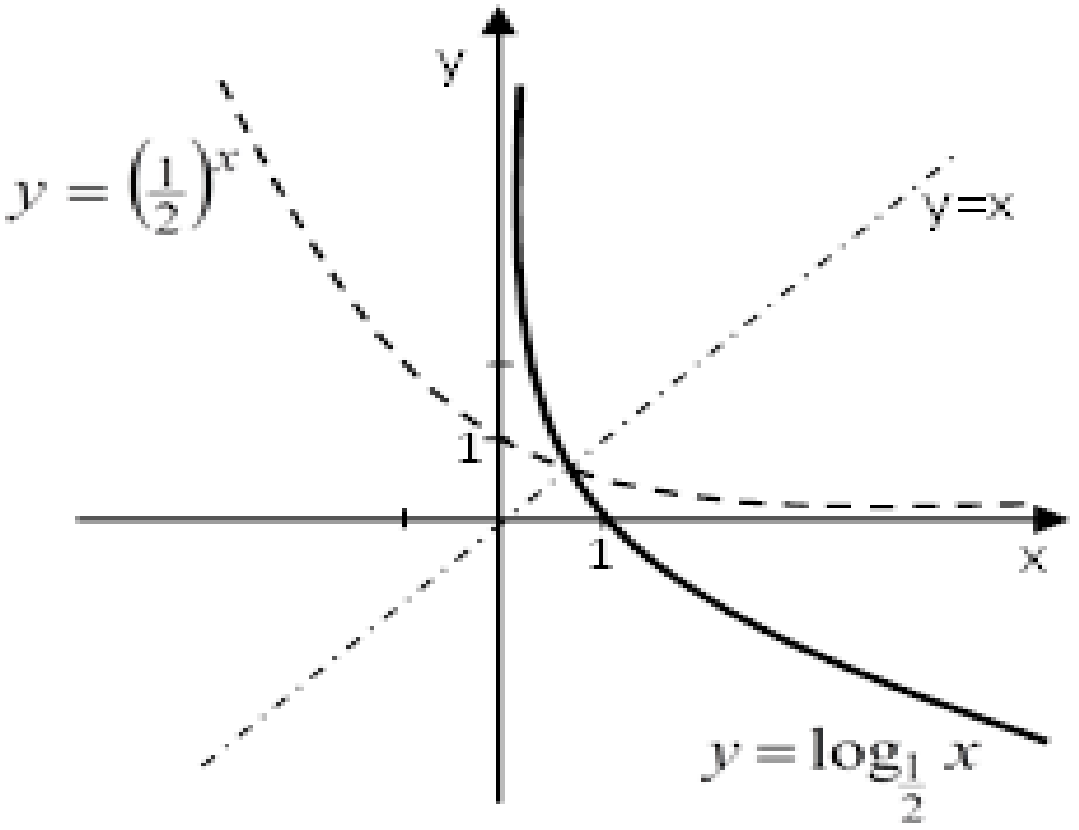
**Example 3.35**: Sketch the graph of the following functions. Find the domain and range of each.

a) $y = \log_3 x$ b) $y = \log_{\frac{1}{2}} x$

**Solution**: a) Since $y = \log_3 x$ is the inverse of $y = 3^x$, we can obtain the graph of $y = \log_3 x$ by reflecting the graph of $y = 3^x$ about the line $y = x$, as shown below.



b) To get the graph of $y = \log_{\frac{1}{2}} x$, we reflect the graph of $y = \left(\frac{1}{2}\right)^x$ about the line $y = x$ as shown below



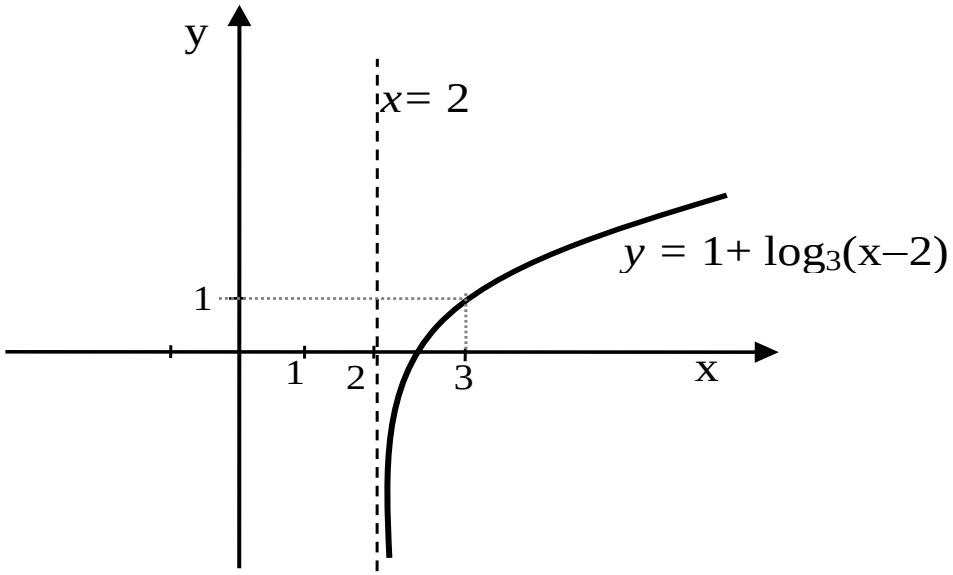

Taking note of the features of the two graphs we have the following important information about the graph of the log function:

| **The Logarithmic Function** $y = \log_b x$ |
|---|
| 1. Its domain is the set of positive real numbers<br>2. Its range is the set of real numbers.<br>3. Its graph exhibits logarithmic growth if $b > 1$ and logarithmic decay if $0 < b < 1$.<br>4. The $x-$ intercept is 1. There is no $y-$ intercept.<br>5. The $y-axis$ is a vertical asymptote. |

**Example 3.36**:

1. Sketch the graph of $f(x) = 1 + \log_3(x-2)$. Find the domain, range, asymptote and intercepts.

   **Solution**: We can obtain the graph of $y = 1 + \log_3(x-2)$ by applying the graphing principle to shift the basic logarithmic growth graph 2 units to the right and 1 unit up.



We have $Dom(f)=\{x: x>2\}$, $Range(f)=\Re$ and the graph has the line $x=2$ as a vertical asymptote. To find the intercept, we set $y=0$ and solve for $x$. Setting $y=0$ and solving for $x$, we will obtain $x=\frac{7}{3}$. Thus, the $x-$intercept is $\frac{7}{3}$.

1. Find the inverse function for

a) $y=f(x)=3^x+4$ b) $y=g(x)=\log_3(x-2)$

**Solution**: Following the procedure for finding an inverse function, we have

(a) $y = 3^x + 4$ Interchange $x$ and $y$

$x = 3^y + 4$ solve explicitly for $y$

$x - 4 = 3^y$ Write in logarithmic form

$y = \log_3(x-4)$

Thus, $f^{-1}(x) = \log_3(x-4)$

(b) $y = \log_3(x-2)$ Interchange $x$ and $y$

$x = \log_3(y-2)$ Write in logarithmic form

$y - 2 = 3^x$ solve explicitly for $y$

$y = 3^x + 2$

Thus, $g^{-1}(x) = 3^x + 2$

The following table contains the basic properties of logarithm:

## Properties of logarithm

Assume that $b, u$ and $v$ are positive and $b \neq 1$. Then

1. $\log_b(uv) = \log_b u + \log_b v$. In words, logarithm of a product is equal to the sum of the logs of the factors.
2. $\log_b\left(\frac{u}{v}\right) = \log_b u - \log_b v$. In words, the log of a quotient is the log of the numerator minus the log of the denominator.
3. $\log_b\left(u^r\right) = r\log_b u$. In words, the log of a power is the exponent times the log.
4. $\log_b(b^x) = x\log_b b = x$
5. $b^{\log_b x} = x$

**Example3.3.7**

1. Express in terms of simpler logarithms.

   a) $\log_b(x^3y)$ b) $\log_b(x^3+y)$ c) $\log_b\left(\frac{\sqrt{xy}}{z^3}\right)$

   **Solution**

   a) $\log_b(x^3y)=\log_b x^3+\log_b y=3\log_b x+\log_b y$

   b) Examining the properties of logarithms, we can see that they deal with log of a product, quotient and power. Thus, $\log_3(x^3+y)$ which is the log of a sum cannot be simplified using log properties.

   c) We have

   $$\log_b\left(\frac{\sqrt{xy}}{z^3}\right)=\log_b\sqrt{xy}-\log_b(z^3)=\log_b(xy)^{\frac{1}{2}}-3\log_b z=\frac{1}{2}(\log_b x+\log_b y)-3\log_b z.$$

2. Show that $\log_b\frac{1}{2}=-\log_b 2$.

   **Solution** We have $\log_b\frac{1}{2}=\log_b 1-\log_b 2=0-\log_b 2=-\log_b 2$.

The logarithmic function was introduced without stressing the particular base chosen. However, there are two bases of special importance in science and mathematics, namely, $b = 10$ and $b = e$

**Definition 3.21: (Common Logarithm)**

$f(x) = \log_{10} x$ is called the common logarithm function. We write $\log_{10} x = \log x$.

The inverse of the natural exponential function is called the natural logarithmic function and has its own special notation.

**Definition 3.22: (Natural Logarithm)**

$f(x) = \log_e x$ is called the natural logarithmic function. We write $\log_e x = \ln x$.

**Example 3.38**:

1. Evaluate $\log 1000$

   **Solution**: Let $a = \log 1000$. Then, $a = \log_{10} 1000 = \log_{10}(10^3) = 3$.

2. Find the inverse function of $f(x) = e^x + 1$.

   **Solution**: Let $y = e^x + 1$ Interchange $x$ and $y$

   $x = e^y + 1$ Solve for $y$

   $x - 1 = e^y$ Rewrite in logarithmic form

   $y = \ln(x - 1)$

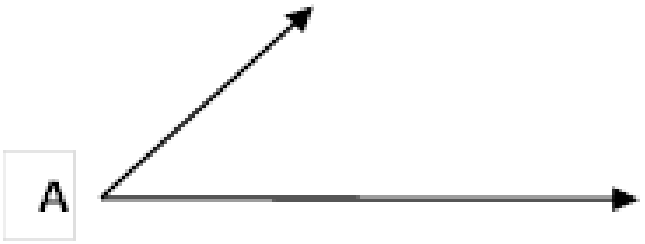
   Thus, $f^{-1}(x) = \ln(x - 1)$.

# Trigonometric functions and their graphs

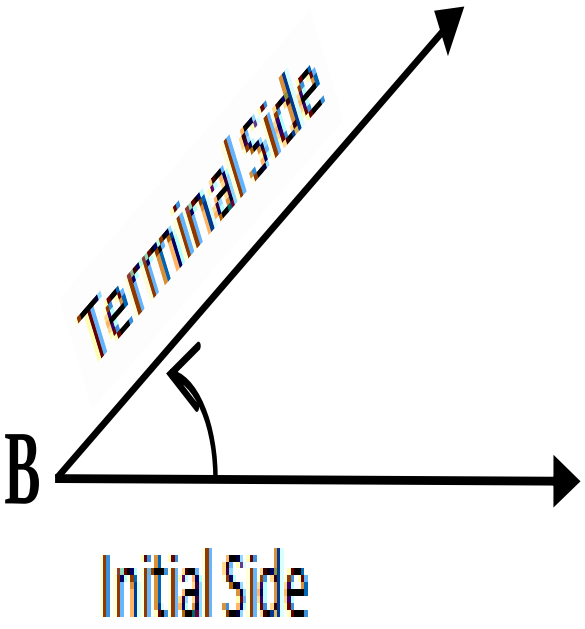
The functions sine, cosine, tangent, cotangent, secant and cosecant are called trigonometric functions. The domain of the trigonometric functions is more naturally the set of all geometric angles.

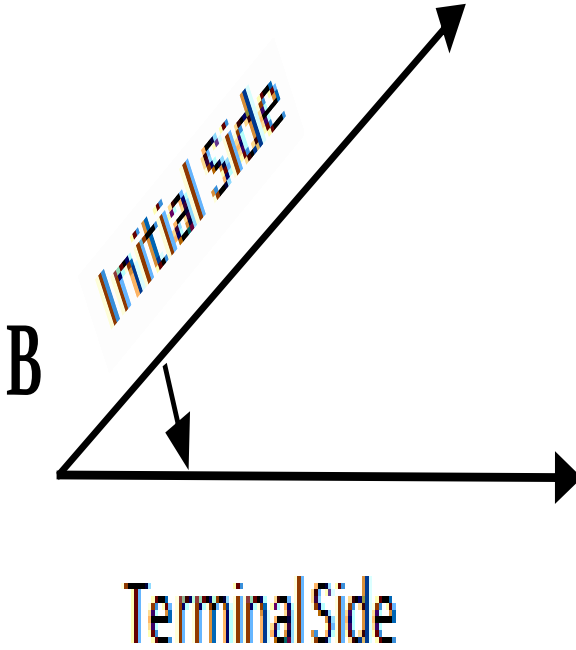
## Angle Measurement

An angle is the figure formed by two half-lines or rays with a common end point. The common end point is called the vertex of the angle.

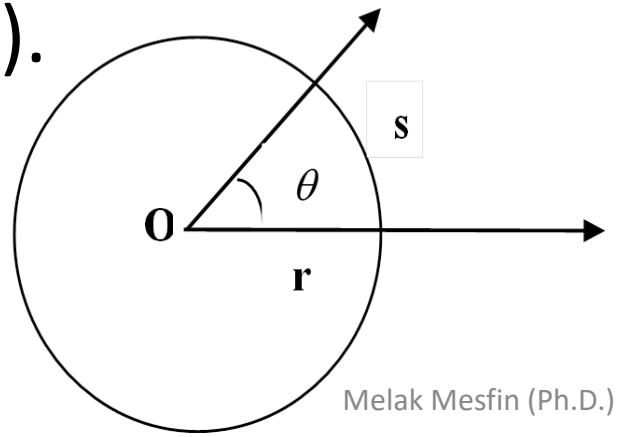


In forming the angle, one side remains fixed and the other side rotates. The fixed side is called the initial side and the side that rotates is called the terminal side. If the terminal side rotates in a counter clockwise direction, we call the angle positive angle, and if the terminal side rotates in a clockwise direction, we call the angle negative angle.



The unit of measurement for angle is **degree** denoted (°). Recall that the measure of a full round angle (full circle) is 360°, straight angle is 180°, and right angle is 90°. An alternative unit measure for angles is the **radian measure**. To see the relation between the degree measure and radian measure of an angle, let us consider an angle and draw a circle of radius  with the vertex of  at its center . Let  represent the length of the arc of the circle intercepted by (as shown below).



From basic geometry we have that;

$$\frac{\theta}{one\ complete\ rotation} = \frac{s}{circumfrence\ of circle}$$

$$= \frac{s}{2\pi r}$$

$$= \frac{r\theta\ in\ radian}{2\pi r}$$

Thus, we have the following conversion formula:

$$\frac{\theta\ in\ degrees}{180^\circ} = \frac{\theta\ in\ radians}{\pi}$$

**Example 3.39**:

1. Convert each of the following radian measures to degrees.

   a) $\frac{\pi}{6}$ b) $\frac{3\pi}{5}$

**Solution**: a) By the conversion formula, we have $\frac{\theta}{180^\circ} = \frac{\frac{\pi}{6}}{\pi}$, which implies that $\theta = 30^\circ$.

b) Again using the conversion formula, we get $\frac{\theta}{180^\circ} = \frac{\frac{3\pi}{5}}{\pi}$, which implies that $\theta = 108^\circ$.

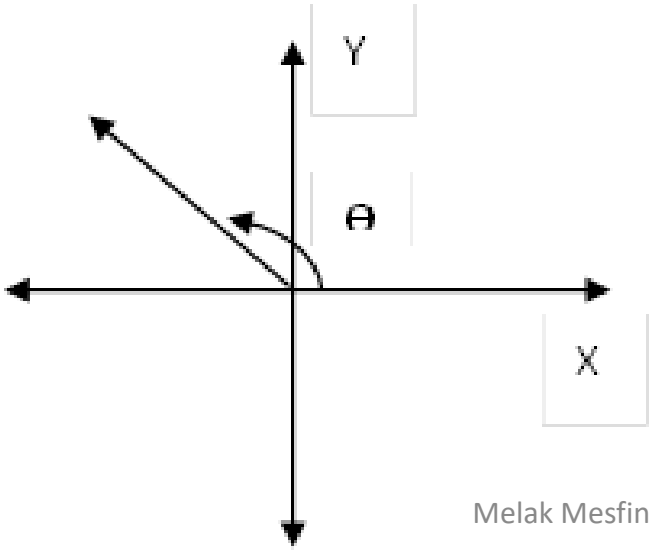
2. Convert to radian measures

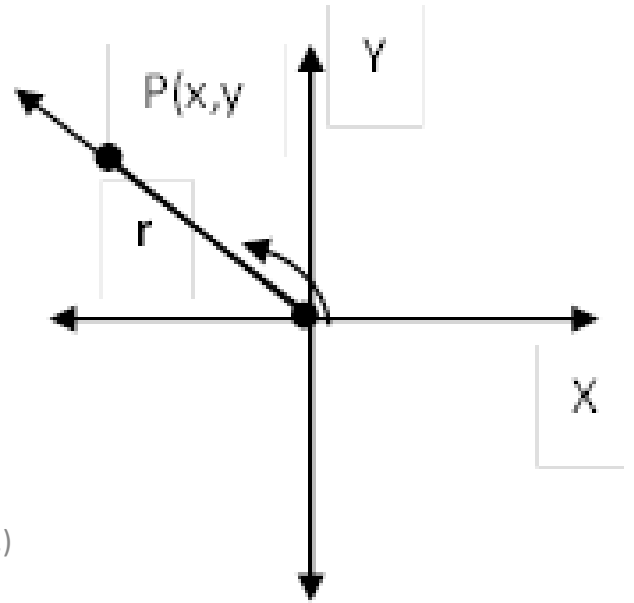
   a) $90^\circ$ b) $270^\circ$

**Solution**: a) Let $\theta$ represent the radian measure of $90^\circ$. Using the conversion formula, we obtain: $\frac{\theta}{\pi} = \frac{90^\circ}{180^\circ}$, which implies that $\theta = \frac{\pi}{2}$.

b) Rather than using the conversion formula, we notice that $270^\circ = 3(90^\circ)$. In part (a) we found that $90^\circ = \frac{\pi}{2}$, and so we have $270^\circ = \frac{3\pi}{2}$.

To define the trigonometric functions, we will view all angles in the context of a Cartesian coordinate system: that is, given an angle $\theta$, we begin by putting $\theta$ in standard position, meaning that the vertex of $\theta$ is placed at the origin and initial side of $\theta$ is placed along the positive x-axis . Thus the location of the terminal side of $\theta$ will, of course, depend on the size of $\theta$.



We then locate a point( other than the origin) on the terminal side of $\theta$ and identify its coordinates $(x, y)$ and its distance to the origin, dented by $r$ . Then, $r$ is positive.

With $\theta$ in standard position, we define the six trigonometric functions of $\theta$ as follows:

**Definition 3.23**

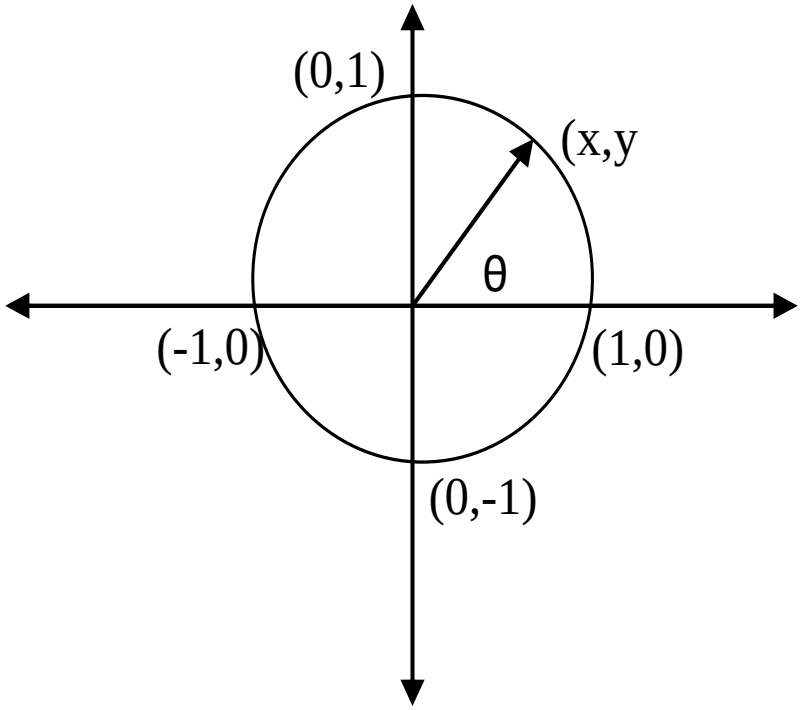
| **Name of function** | **Abbreviation** | **Definition** |
|---|---|---|
| Sine $\theta$ | $\sin\theta$ | $\sin\theta = \frac{y}{r}$ |
| Cosine $\theta$ | $\cos\theta$ | $\cos\theta = \frac{x}{r}$ |
| Tangent $\theta$ | $\tan\theta$ | $\tan\theta = \frac{y}{x}$ |
| Cosecant $\theta$ | $\csc\theta$ | $\csc\theta = \frac{r}{y}$ |
| Secant $\theta$ | $\sec\theta$ | $\sec\theta = \frac{r}{x}$ |
| Cotangent $\theta$ | $\cot\theta$ | $\cot\theta = \frac{x}{y}$ |

Recall that the radian measure of an angle is defined as $\theta = \frac{s}{r}$, where $\theta$ is angle in radians, $s$ is the length of the arc intercepted by $\theta$ and $r$ is the length of the radius. Since $s$ and $r$ are both lengths, the quotient $\frac{s}{r}$ is a pure number without any units attached. Thus, any angle can be interpreted as a real number. Conversely, any real number can be interpreted as an angle. Thus, we can describe the domains of the trigonometric functions in the frame work of the real number systems. If we let $f(\theta) = sin\theta$, the domain consists of all real numbers $\theta$ for which $sin\theta$ is defined. Since $sin\theta = \frac{y}{r}$ and $r$ is never equal to zero, the domain for $sin\theta$ is the set of all real numbers. Similarly, the domain of $f(\theta) = cos\theta = \frac{x}{r}$ is also the set of all real numbers.

- **The graph of** $y = \sin\theta$

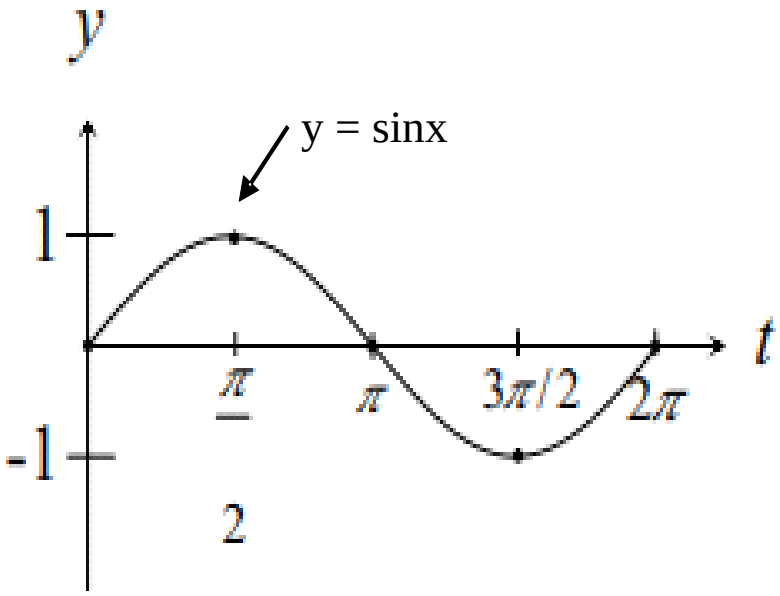
To analyze $f(\theta) = \sin\theta$, we keep in mind that once we choose a real number $\theta$, we draw the angle, in standard position, that corresponds to $\theta$. To simplify our analysis, we choose the point $(x, y)$ on the terminal side so that $r = 1$. That is, $(x, y)$ is a point on the unit circle $x^2 + y^2 = 1$.

Note that $\sin\theta = \frac{y}{1} = y$.

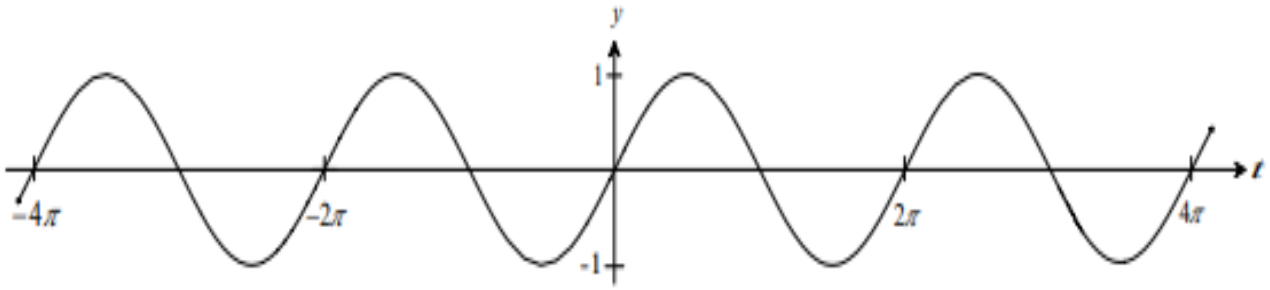


1. As $\theta$ increases from 0 to $\frac{\pi}{2}$, $y = sin\theta$ increases from 0 to 1.
2. As $\theta$ increases from $\frac{\pi}{2}$ to $\pi$ , $y = sin\theta$ decreases from 1 to 0.
3. As $\theta$ increases from $\pi$ to $\frac{3\pi}{2}$ ,$y = sin\theta$ decreases from 0 to -1.
4. As $\theta$ increases from $\frac{3\pi}{2}$ to $2\pi$ , $y = sin\theta$ increases from -1 to 0.

Based on this analysis, we have the graph of $y = sin\theta$ as below:



Since the values of $f(\theta) = \sin\theta$ depend only on the position of the terminal side, adding or subtracting multiples of $2\pi$ to $\theta$ will leave the value of $f(\theta) = \sin\theta$ unchanged. Thus, the values of $f(\theta) = \sin\theta$ will repeat every $2\pi$ units. The complete graph of $f(\theta) = \sin\theta$ appears below.




The graph of $y = \sin x$, which is called the basic sine curve.

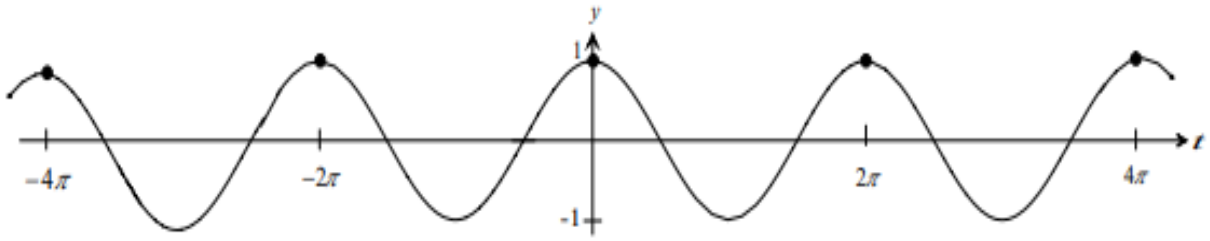
- **The graph of** $y = \cos\theta$

Applying the same type of analysis to $f(\theta) = \cos\theta$, we will able to get a good idea of what its graph looks like. The figure below shows the angle corresponding to $\theta$ as it increases through quadrant I, II, III and IV.

Keeping in mind that $\cos\theta = \frac{x}{1} = x$, we have the following:

1. As $\theta$ increases from 0 to $\frac{\pi}{2}$, $x = \cos\theta$ decreases from 1 to 0.
2. As $\theta$ increases from $\frac{\pi}{2}$ to $\pi$, $x = \cos\theta$ decreases from 0 to – 1.
3. As $\theta$ increases from $\pi$ to $\frac{3\pi}{2}$, $x = \cos\theta$ increases from – 1 to 0.
4. As $\theta$ increases from $\frac{3\pi}{2}$ to $2\pi$, $x = \cos\theta$ increases from 0 to 1.

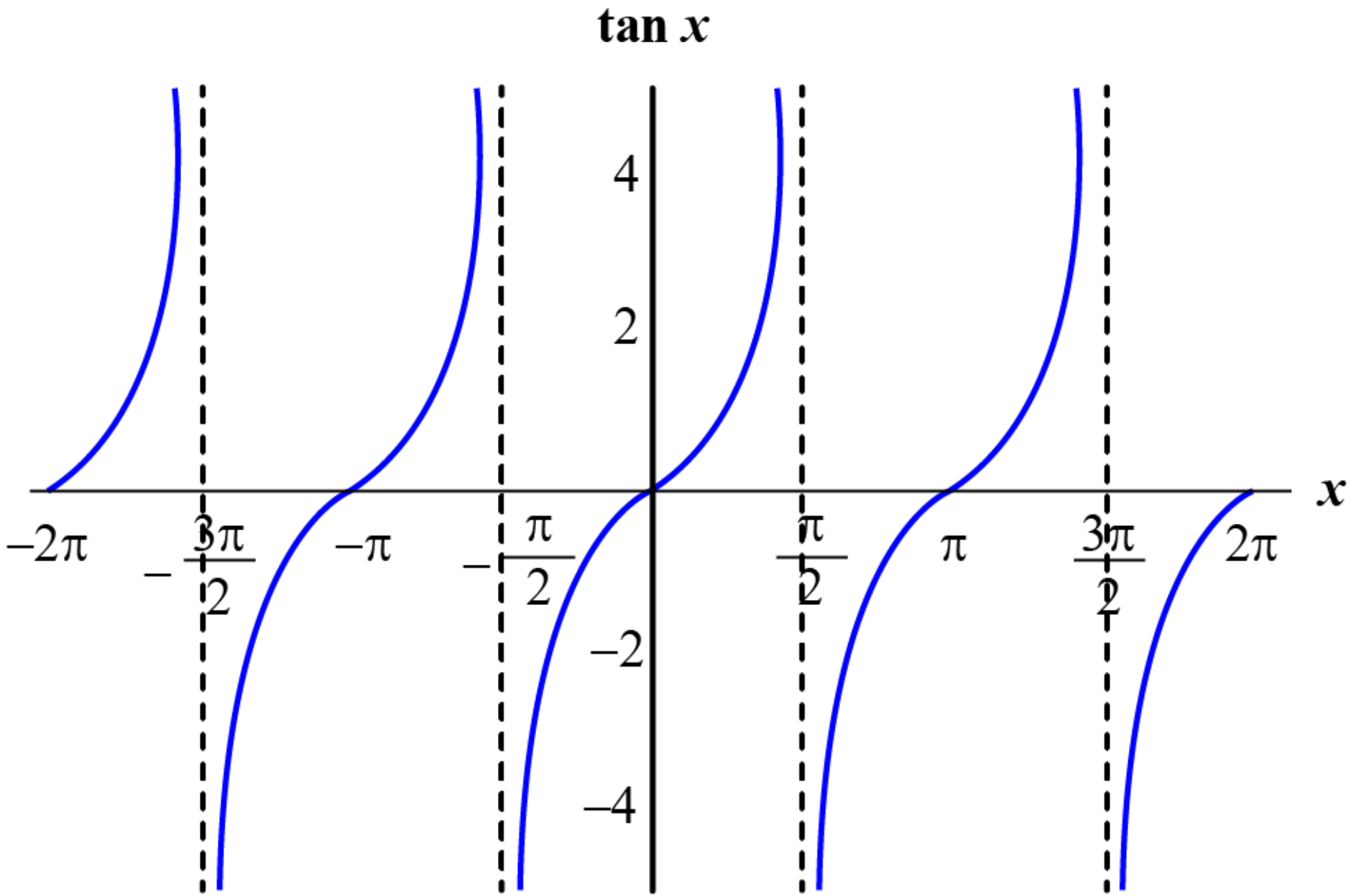
Based on this analysis, we have the graph of $f(\theta) = \cos\theta$ as shown below:




- **The graph of** $y = \tan\theta$

Since $\tan\theta = \frac{y}{x}$ is undefined whenever $x = 0$, $\tan\theta$ is undefined whenever the terminal side of the angle corresponding to $\theta$ falls on the $y-axis$. This happens for $\theta = \frac{\pi}{2}$, to which we can add or subtract any multiple of $\pi$ that will again bring the terminal side back to the $y-axis$. Thus, domain of $\tan\theta$ is $\{\theta : \theta \neq \frac{\pi}{2} + n\pi\}$, where $n$ is an integer.

| $\theta$ | $0°$ | $\frac{\pi}{2}$ | $\pi$ | $\frac{3\pi}{2}$ | $2\pi$ |
|---|---|---|---|---|---|
| x | 1 | 0 | -1 | 0 | 1 |
| y | 0 | 1 | 0 | -1 | 0 |
| $\tan\theta$ | 0 | $\infty$ | 0 | $-\infty$ | 0 |


tan x
4
2
−2
−4
x
−2π
−3π/2
−π
−π/2
π/2
π
3π/2
2π

**Definition 3.24: (Periodic Function)**

A function $y = f(x)$ is called periodic if there exists a number $p$ such that $f(x+p) = f(x)$ for all $x$ in the domain of $f$. The smallest such number $p$ is called the period of the function

A periodic function keeps repeating the same set of $y-values$ over and over again. The graph of a periodic function shows the same basic segment of its graph being repeated. In the case of sine and cosine functions, the period is $2\pi$. The period of the tangent function is $\pi$.

**Definition 3.25: (Amplitude of a periodic function)**

The amplitude of a periodic function $f(x)$ is

$$A = \frac{1}{2}[\text{ maximum value of } f(x) - \text{minimum value of } f(x)]$$

Thus, the amplitude of the basic sine and cosine function is 1.

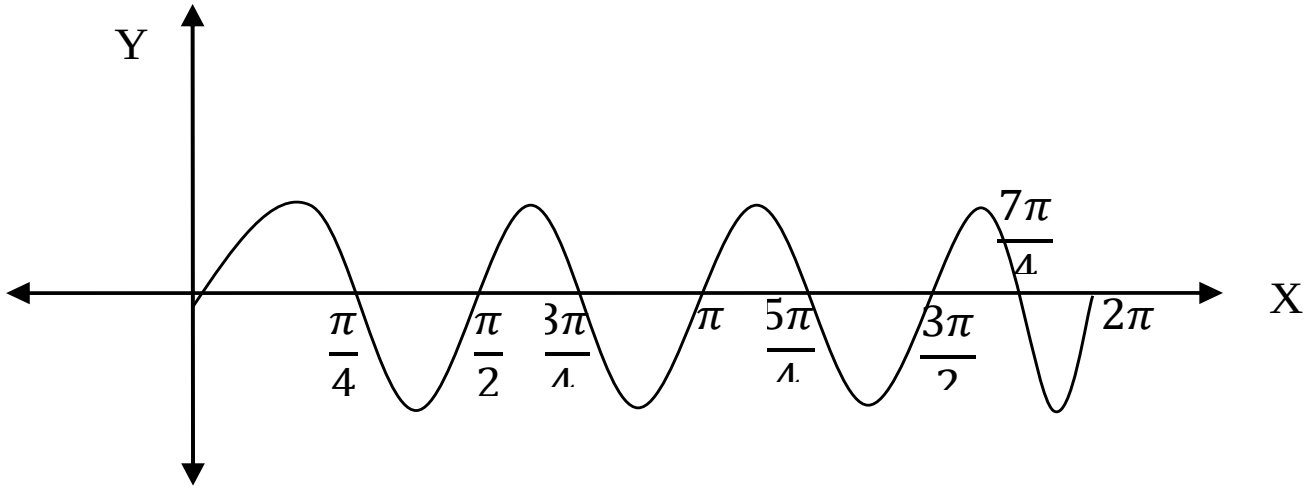
The portion of the graph of a sine or cosine function over one period is called a complete cycle of the graph. In other words, the minimal portion of a sine or cosine graph that keeps repeating itself is called a complete cycle of the graph.

**Definition 3.26: (Frequency of a periodic function)**

The number of complete cycles a sine or cosine graph makes on an interval of length equal to $2\pi$ is called its frequency.

The frequency of the basic sine curve $y = \sin x$ and the basic cosine curve $y = \cos x$ is 1, because each graph makes 1 complete cycle in the interval $[0, 2\pi]$.

If a sine function has period of $\frac{\pi}{2}$ (see the figure below), then the number of complete cycles its graph will make in an interval of length $2\pi$ is $\frac{2\pi}{\pi/2} = 4$.




A sine graph of period $\frac{\pi}{2}$ and frequency 4

**Example3.40**: Sketch the graph of $y = \sin 2x$ and find its amplitude, period and frequency.

**Solution**: We can obtain this graph by applying our knowledge of the basic sine graph. For the basic curve, we have

$$\sin 0 = 0 \qquad \sin\tfrac{\pi}{2} = 1 \qquad \sin\pi = 0 \qquad \sin\tfrac{3\pi}{2} = -1 \qquad \sin 2\pi = 0$$

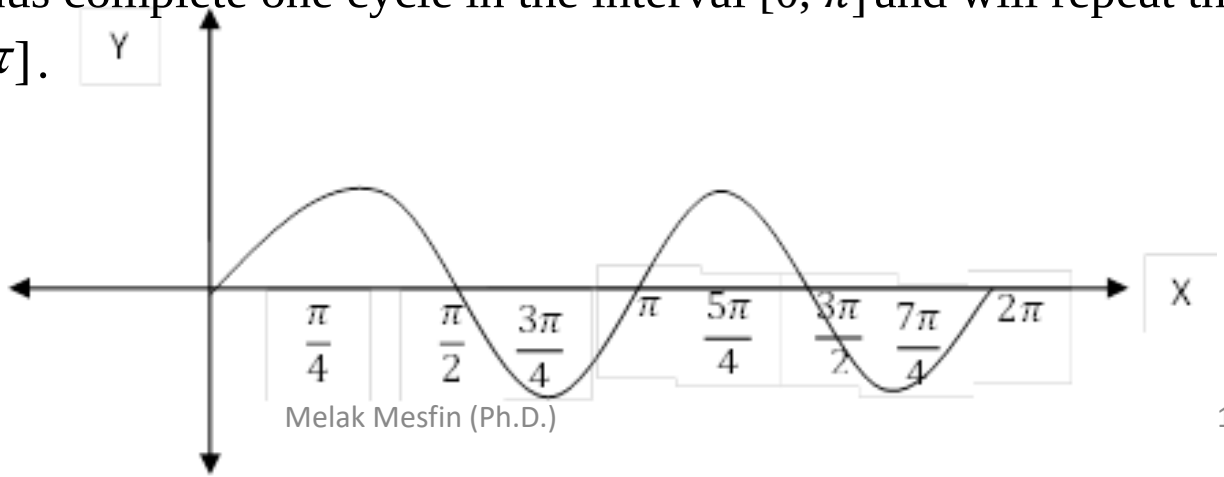
These quadrantal values serve as guide points, which help us draw the graph. To obtain similar guide points for $y = \sin 2x$, we ask for what values of $x$ is

$$2x = 0 \qquad 2x = \tfrac{\pi}{2} \qquad 2x = \pi \qquad 2x = \tfrac{3\pi}{2} \qquad 2x = 2\pi$$

and we get

$$x = 0 \qquad x = \tfrac{\pi}{4} \qquad x = \tfrac{\pi}{2} \qquad x = \tfrac{3\pi}{4} \qquad x = \pi$$

Thus, $y = \sin 2x$ will have the values 0, 1, 0, $-1$, 0 at $x = 0, \tfrac{\pi}{4}, \tfrac{\pi}{2}, \tfrac{3\pi}{4},$ and $\pi$, respectively. The graph of $y = \sin 2x$ will thus complete one cycle in the interval $[0, \pi]$ and will repeat the same values in the interval $[\pi, 2\pi]$.



From this graph we see that $y = \sin 2x$ has an amplitude of 1, a period $\pi$, and a frequency of 2.

For convenience we summarize our discussion on the domains of the trigonometric functions in the table.

| | |
|---|---|
| 1. $f(x) = \sin x$ | Domain = All real numbers |
| 2. $f(x) = \cos x$ | Domain = All real numbers |
| 3. $f(x) = \tan x$ | Domain = $\{x : x \neq \frac{\pi}{2} + n\pi\}$ |
| 4. $f(x) = \csc x$ | Domain = $\{ x : x \neq n\pi \}$ |
| 5. $f(x) = \sec x$ | Domain = $\{x : x \neq \frac{\pi}{2} + n\pi\}$ |
| 6. $f(x) = \cot x$ | Domain = $\{x : x \neq n\pi\}$ |
| | where $n$ is an integer |

We have the following trigonometric identities

1. $\sin^2 x + \cos^2 x = 1$
2. $\tan^2 x + 1 = sce^2 x$
3. $1 + \cot^2 x = \csc^2 x$

**Note:** We know that the period of sin(ax) is p= 2π/a and the fundamental period of cos(bx) is p=2π/b.

- **Hyperbolic functions and their graphs**

The hyperbolic functions are certain combinations of exponential functions, that occur in various applications, with properties similar to those of the trigonometric functions. Among many other applications they are used to describe the formation of satellite rings around the planets, to describe the shape of a rope hanging from two points, and have application in relativity theory. The two basic hyperbolic functions are the hyperbolic sine and hyperbolic cosine functions. They are defined as follows.

**Definition 3.27**:

1. The **hyperbolic sine** function is defined by:

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

The domain of $\sinh x$ is $\Re$.

2. The **hyperbolic cosine** function is defined by:

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

The domain of $\cosh x$ is also $\Re$.

**Remark**

1. $\cosh x$ is pronounced "*kosh*" $x$ and $\sinh x$ is pronounced as "*sinch*" $x$.
2. Since $e^{-x} > 0$ for all $x \in \Re$, we see that $\cosh x > \sinh x$ for every $x \in \Re$.
3. If $f(x) = \dfrac{e^x + e^{-x}}{2}$, then $f(-x) = \dfrac{e^{-x} + e^x}{2} = \dfrac{e^x + e^{-x}}{2} = f(x)$. Thus $\cosh x$ is an even function.
4. *Sinhx* is an odd function.
5. In contrast to sine and cosine, the hyperbolic functions are not periodic.

**Example 3.28:** Using the above definitions, show that

1. $\cosh^2 x - \sinh^2 x = 1$
2. $\sinh(x+y) = \cosh x \sinh y + \sinh x \cosh y$
3. $\cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$

**Solution**:

1. We have

$$\left(\frac{e^x+e^{-x}}{2}\right)^2-\left(\frac{e^x-e^{-x}}{2}\right)^2=\left(\frac{e^{2x}+2+e^{-2x}}{4}\right)-\left(\frac{e^{2x}-2+e^{-2x}}{4}\right)=1$$

2. $$\sinh(x+y)=\frac{e^{x+y}-e^{-x-y}}{2}=\frac{e^xe^y-e^{-x}e^{-y}}{2}=\frac{2e^xe^y-2e^{-x}e^{-y}}{4}$$

$$=\frac{e^xe^y-e^xe^{-y}+e^{-x}e^y-e^{-x}e^{-y}}{4}+\frac{e^xe^y+e^xe^{-y}-e^{-x}e^y-e^{-x-y}}{4}$$

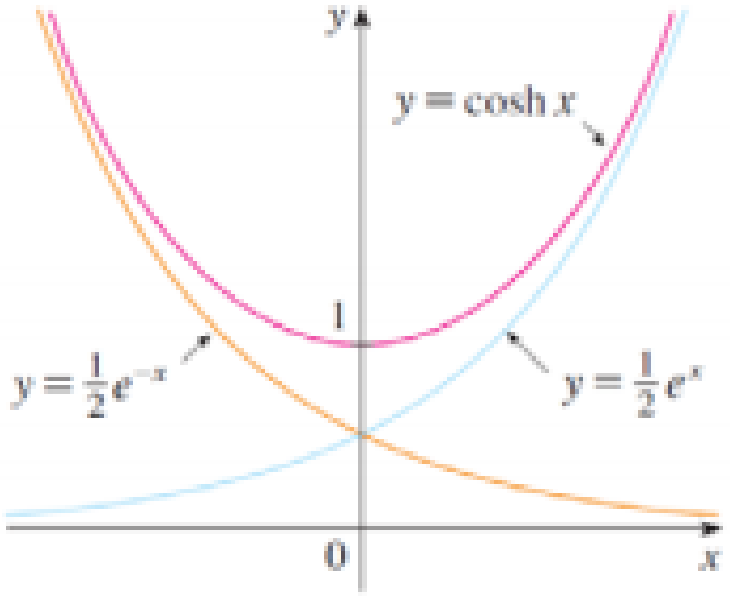
$$=\left(\frac{e^x+e^{-x}}{2}\right)\left(\frac{e^y-e^{-y}}{2}\right)+\left(\frac{e^x-e^{-x}}{2}\right)\left(\frac{e^y+e^{-y}}{2}\right)$$

$$=\cosh x\sinh y+\sinh x\cosh y$$

3. Similar

- **The graph of** $y = \cosh x$

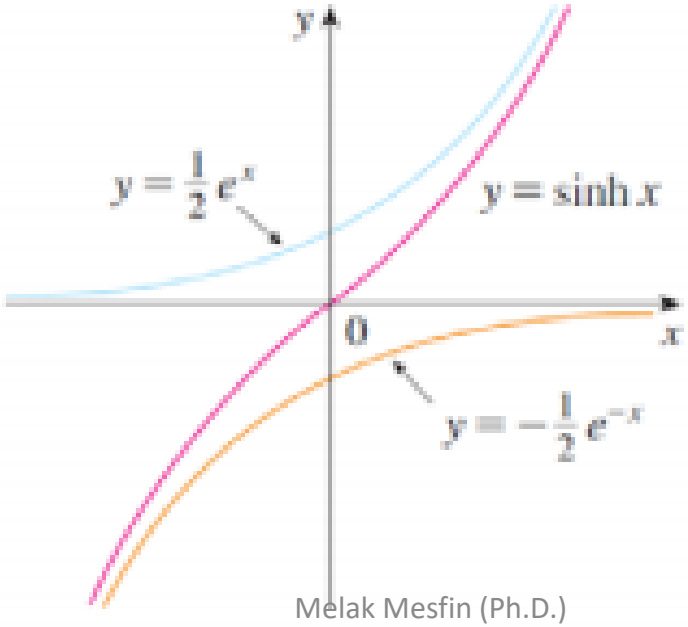
  Since $\cosh x$ is an even function, its graph is symmetric about the $y-axis$. Its $y-$ intercept is $(0,1)$, because $\cosh(0) = 1$. As $x$ tends to infinity, $\cosh x = \frac{e^x}{2} + \frac{e^{-x}}{2}$ tends to infinity because $\frac{e^x}{2}$ goes to infinity and $\frac{e^{-x}}{2}$ approaches to 0. When $x$ is a large negative number $\cosh x$ acts like $\frac{e^{-x}}{2}$, because $\frac{e^x}{2}$ gets close to 0. Thus the graph of $y = \cosh x$ looks like:



This graph can also be obtained by geometrically adding the two curves $y = e^x$ and $y = e^{-x}$, and taking half of each resulting $y - value$. Observe that range of $\cosh x$ is $[1, \infty)$.

- **The graph of** $y = \sinh x$

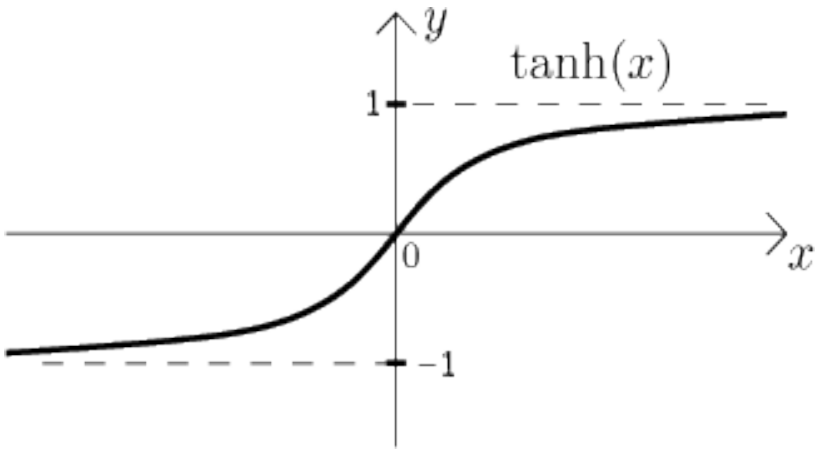
Since $\sinh x$ is an odd function, its graph is symmetric about the origin. The graph passes through the origin because $\sinh(0) = 0$. As $x$ gets large $\sinh x$ acts like $\frac{e^x}{2}$ and when $x$ is a large negative number, $\sinh x$ acts like $-\frac{e^{-x}}{2}$. Thus, the graph of $y = \sinh x$ looks like:

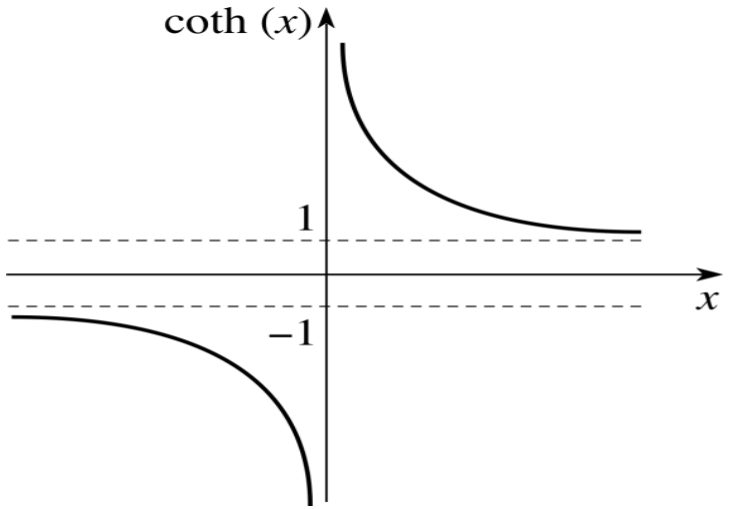


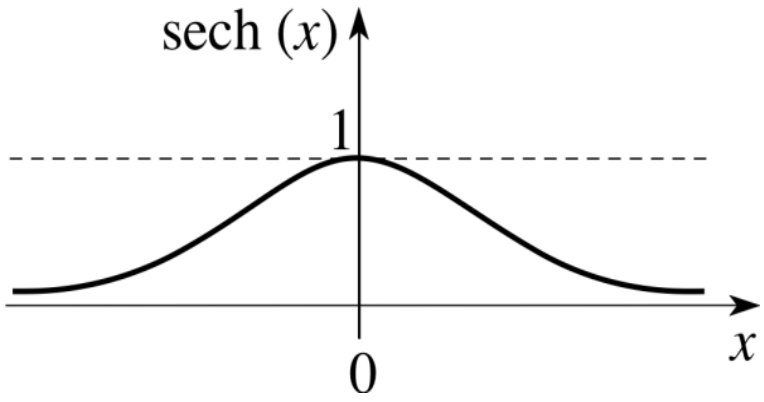
The remaining four hyperbolic functions are defined in terms of $\cosh x$ and $\sinh x$ by analogy with trigonometry.

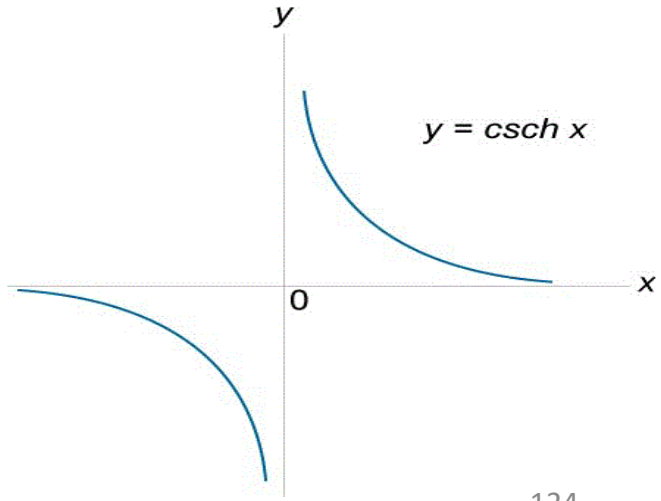
| | |
|---|---|
| $\tanh x = \dfrac{\sinh x}{\cosh x} = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$ | (The domain of $\tanh x$ is $\Re$) . |
| $\coth x = \dfrac{\cosh x}{\sinh x} = \dfrac{e^x + e^{-x}}{e^x - e^{-x}}$ | (The domain of $\coth x$ is $\Re \setminus \{0\}$ ) |
| $\sec h\, x = \dfrac{1}{\cosh x} = \dfrac{2}{e^x + e^{-x}}$ | (The domain of $\sec hx$ is $\Re$ ) |
| $\csc h\, x = \dfrac{1}{\sinh x} = \dfrac{2}{e^x - e^{-x}}$ | (The domain of $\csc h\, x$ is $\Re \setminus \{0\}$ ) |

The graphs of these four functions are given below:


y
tanh(x)
1
0
x
−1


coth (x)
1
x
−1


sech (x)
1
0
x


y
y = csch x
0
x

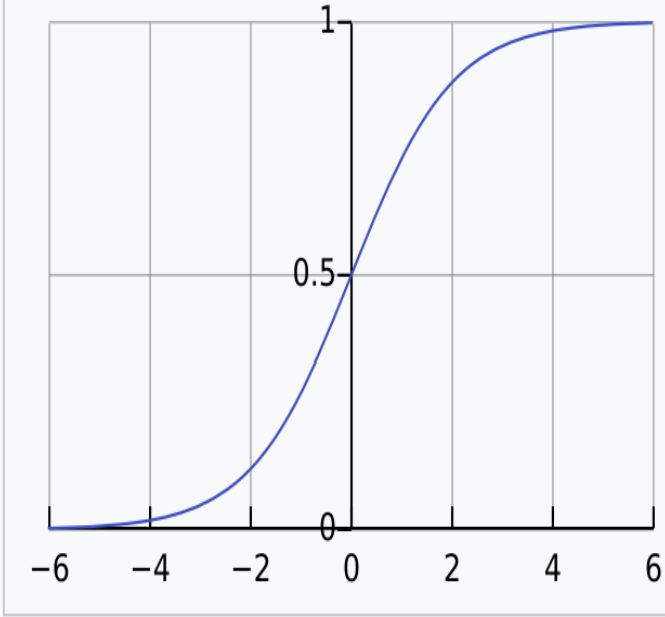
# Example

A **sigmoid function** is any mathematical function whose graph has a characteristic S-shaped or **sigmoid curve**.

A common example of a sigmoid function is the logistic function, which is defined by the formula:[1]

$$\sigma(x) = \frac{1}{1+e^{-x}} = \frac{e^x}{1+e^x} = 1 - \sigma(-x).$$

Other sigmoid functions are given in the Examples section. In some fields, most notably in the context of artificial neural networks, the term "sigmoid function" is used as a synonym for "logistic function".




The logistic curve

The trigonometric terminology and notation for the hyperbolic functions stem from the fact that they satisfy a list of identities that much resemble the familiar trigonometric identities, apart from an occasional difference of sign.

$$\cosh^2 x - \sinh^2 x = 1 \qquad (1)$$

$$1 - \tanh^2 x = \operatorname{sec} h^2 x \qquad (2)$$

$$\coth^2 x - 1 = \operatorname{csc} h^2 x \qquad (3)$$

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y \qquad (4)$$

$$\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y \qquad (5)$$

The trigonometric functions are sometimes called circular functions because the point $(\cos\theta, \sin\theta)$ lies on the circle $x^2 + y^2 = 1$ for all $\theta$. Similarly, identity (1) tells us that the point $(\cosh\theta, \sinh\theta)$ lies on the hyperbola $x^2 - y^2 = 1$, and this is the reason for the name hyperbolic functions.